A Cascade Domain Clustering Algorithm for Multiview DSM Fusion From Urban Satellite Images

Li Zhao, Yubo Men, Yi Zhu, Haiyan Wang, and Chaoguang Men

***Abstract*— Multiview digital surface model (DSM) fusion has emerged as an important technique for 3-D reconstruction of multiview satellite images. However, existing multiview DSM fusion approaches are prone to the problems of blurred elevation divisions at the object edges, salt and pepper noises on smooth surfaces, and severe loss of surface details in weakly textured regions. In this article, we present a cascade domain clustering (CDC) algorithm for fusing multiview DSMs, which is realized by the combination of salient domain clustering and model domain clustering. Initially, the salient domain clustering is employed to demarcate prominent objects and identify regional edges using 2-D spectral and 3-D elevation information. Subsequently, to further segment the intricate surface structures, particularly for objects with low texture attributes, we implement the model domain clustering to iteratively aggregate 3-D points and fit geometric models corresponding to the aggregated clusters. Finally, the multiview DSMs are fused iteratively through the weighted least-squares (WLS) method, with model clusters serving as the fundamental units, under the constraints of the geometric models. Sufficient experiments show that the proposed CDC algorithm surpasses other popular multiview DSM fusion algorithms in terms of completeness and accuracy, achieving 91.51% completeness and 0.93-m root-mean-square error (RMSE), representing a 79.89% improvement in completeness and an 86.40% reduction in RMSE compared to the popular stereo 3-D reconstruction pipeline.**

***Index Terms*— Cascade domain clustering (CDC), geometric model, multiview digital surface model (DSM), urban satellite image.**

I. Introduction

AUTOMATIC 3-D reconstruction of large-scale scenes has been an active area of research in photogrammetry and computer vision. The state-of-the-art stereo 3-D reconstruction approaches have achieved country-wide reconstructions utilizing high-resolution satellite imageries with submeter ground sampling distance (GSD) [1]. Furthermore, the proposed pipelines for the stereo 3-D reconstruction of satellite images typically encompass stereo rectification, stereo matching, triangulation based on rational function modeling (RFM), and subsequent postprocessing [2], [3]. Improvements for stereo 3-D reconstruction of satellite images have also mostly focused on stereo matching, due to the fact that the disparity maps obtained in this stage can maintain a tight correspondence with the digital surface model (DSM) [4], [5]. As a result, deep learning-based stereo matching methods, which are ranked high in stereo matching benchmark reviews, are often used in satellite imagery as well. Previously, deep learning-based stereo matching methods have achieved good results by replacing one or more modules in traditional methods (e.g., feature extraction) with deep learning methods [6], [7]. Subsequently, end-to-end algorithms for the whole link are proposed so that various optimization strategies for deep learning can be applied and computational speed and accuracy are improved [8], [9]. Drawing upon this framework, Chen et al. [10] employed a deep residual network to extract perceptual features, which were then used to obtain the 3-D grid coordinates of the object. Subsequently, they calculated the true 3-D coordinates of the target from the satellite parameters. Building upon this foundation, Gao et al. [11] expanded the existing multiview stereo network (MVSNet) by incorporating rational polynomial coefficients (RPCs) correction and multiscale feature extraction from satellite images into the network. This led to the development of a deep learning network based on satellite images, known as Sat-MVSNet.

Nevertheless, the generalization ability and interpretability of the trained model still necessitate verification due to the absence of ground-truth 3-D data utilized for training and the variations in optical sensors. In this manner, exploring more general methods to address elevation errors in weakly textured regions, depth discontinuities, and occluded regions remains a valuable research pursuit. Incorporating DSMs from multiple viewpoints is a straightforward and effective strategy to compensate for occlusions and optimize elevation accuracy [12]. However, the majority of existing DSM fusing methodologies primarily concentrate on the selection of elevations at individual horizontal locations, while neglecting constraints related to object surface structure and geometric characteristics. The fused DSM is susceptible to salt-and-pepper noise on smooth

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surfaces [see Fig. 1(b) and (c)], elevation discontinuities and spikes near height jumps [see Fig. 1(d) and (f)–(h)], and loss of detailed structures [see Fig. 1(e)].

It is a fact that shapes, especially those in urban environments with many man-made structures, often exhibit strong geometric features within a single perceived part. Furthermore, geometric models can play a facilitating role in the 3-D reconstruction of urban scenes, and in scenes with high building density, the geometric properties of a shape play a crucial role in delineating building structures and part boundaries [17]. In single-view reconstruction research, there is a tendency to obtain structure-aware shapes through joint learning of shape geometry and structure, ultimately leading to more accurate single-view 3-D reconstruction results [18], [19]. Nevertheless, existing 3-D reconstruction of satellite imagery estimate 3-D information merely by utilizing stereoscopic techniques and photogrammetric methods since there is no large-scale 3-D shape dataset for learning of shape geometry and structure.

In this article, to optimize the 3-D reconstruction of multiview satellite images, we present a cascade domain clustering (CDC) algorithm for multiview DSM fusion, which aggregates regional and geometric models through unsupervised clustering. Given the large number and complex shapes of DSMs in urban scenarios, the proposed clustering algorithm uses a cascaded approach with salient domain clustering followed by model domain clustering. The salient domain clustering functions as the primary clustering mechanism in the process, categorizing notable regions through the combined utilization of satellite images and their corresponding DSMs. This allows object boundaries with significant elevation drops to be localized by region edges and allows the number of fine surface structures within regions to be reduced. Furthermore, model domain clustering serves as fine clustering that effectively partitions intricate structures within prominent regions, particularly for objects with low-texture attributes. This is achieved through iteratively aggregating 3-D points with analogous geometric features and fitting geometric models corresponding to the identified clusters. Ultimately, the DSMs from multiple views are fused in terms of model clusters under the constraint of the geometric models through weighted least squares (WLS).

Unlike traditional clustering-based DSM fusion approach [2], CDC algorithm does not select directly correct elevations by means of clustering. Instead, it employs clustering to locate dramatic edges and predict regional models. These regional models can serve as positive guides when computing true elevations, rather than directly determining height selection. Additionally, compared to edge constraint-based method [14], CDC refrains from directly utilizing object edges as constraints for DSM fusion. Rather, it employs clustering to group together fine objects and their corresponding models within the salient objects. These clustered groups can serve as the basic units for DSM fusion, avoiding potential misinterpretation caused by faint edges.

Qualitative and quantitative experiments conducted on WorldView-2 (WV-2) and WorldView-3 (WV-3) satellite images demonstrate the superior performance of the proposed method in terms of completeness and accuracy compared to widely used fusion methods. Additionally, the precision and effectiveness of the proposed CDC algorithm for optimizing DSM fusion are validated.

In conclusion, this article makes the following key contributions.

1) Constructs a general DSM fusion framework for 3-D reconstruction of multiview satellite images. The framework is sensor-agnostic, allowing integration of data from any satellite platform, and can serve as an extension to existing stereo 3-D reconstruction approaches by enabling multiview DSM fusion.
2) Presents the novel CDC algorithm to unsupervised segment and fit 3-D geometric models of urban scenes from satellite images. This algorithm effectively aggregates and segments fine ground-level objects by leveraging a synergistic combination of 2-D spectral information and 3-D geometric information.
3) Proposes a multiview DSM iterative fusion algorithm based on WLS, which autonomously predicts accurate elevations by minimizing the residual function (RF).

II. Related Work

In the automatic 3-D reconstruction based on optical images, two basic approaches for acquiring 3-D points are structure from motion (SfM)-based methods and dense matching (DM)-based methods [20]. In comparison to SfM-based methods, DM-based approaches can efficiently generate dense depth maps of expansive scenes within a brief timeframe, which explains why the majority of 3-D reconstruction for high-resolution satellite images chooses to adopt DM-based methods [21]. However, DM-based methods rely on perfect image correction, and current onboard photography techniques make it difficult to obtain well-configured satellite stereo pairs for DM. Such that, there are illumination differences, ground geometric differences, and seasonal differences between satellite stereo pairs. Therefore, there appears to be significant space for improvement in terms of the accuracy and completeness of 3-D reconstruction based on satellite images.

A common optimization method for 3-D reconstruction of stereo satellite images is to provide additional edge constraints in the step of DM by edge detection or region segmentation methods, so that precise disparity maps can be work out [4], [22], [23]. Unfortunately, most stereo 3-D reconstructions still adopt an inferential approach to the elevation calculation of the occlusions, which is essentially *making something out of nothing*, for that the occluded area is the unknowable area. Consequently, the 3-D reconstruction based on stereo satellite images appears to be ineffective in acquiring unbroken DSM in occluded areas. Multiview 3-D reconstruction is currently the most effective means as far as filling elevation holes caused by visual occlusion. According to the way of establishing multiview correspondences, the multiview 3-D reconstruction is often divided into multiview matching (MVM) methods and multistereo matching (MSM) methods [14].

The simultaneous determination of corresponding points across multiple views is essential for MVM, facilitating the establishment of connections between different perspectives

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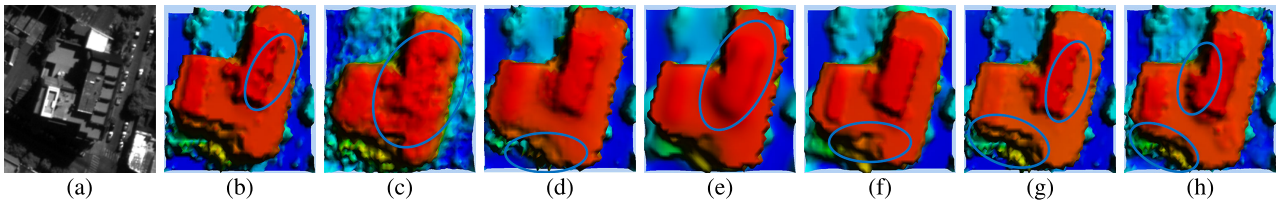


Fig. 1. DSM fusion results for a single building from the state-of-the-art methods. (a) Input image. (b) DSM from probabilistic volume reconstruction method (denoted as PVR15) [13]. (c) DSM from mean filtering method (denoted as MFM15, which is an improved method based on PVR15 by replacing pointwise median calculation with the mean calculation). (d) DSM from adaptive k-medians clustering approach (denoted as AKC17) [2]. (e) DSM from adaptive median filtering method (denoted as AMF19) [14]. (f) DSM from uncertainty-guided depth fusion method (denoted as UGD22) [15]. (g) DSM from adaptive and image-guided fusion method (denoted as AIG22) [16]. (h) DSM from iterative bilateral integration method (denoted as IBI23) [12].

through the assessment of photometric consistency and global visibility. Therefore, MVM is stronger at handling multiview images taken from vehicle and airborne platforms, which have larger baselines and parallaxes than MSM [24]. Moreover, in the step of generating 3-D models, MVM generates more complete point clouds compared with MSM by various fusion methods, including voxel-based methods [25], surface evolution-based methods [26], and interpolation-based growth methods [27]. Among these methods, the voxel-based methods are only applicable to small compact objects in tightly closed spaces. The last two methods rely on reliable initial ground control points (GCPs), which are difficult to obtain on large-scale satellite images.

The MSM is a direct extension of stereo vision, which obtains 3-D point clouds or DSMs by fusing the stereo reconstructed results. Thus, the MSM can directly take advantage of the state-of-the-art DM methods. Compared to MVM, the MSM has been more widely used in large-scale images captured by satellite sensors since they are free from the limitation of GCPs [28]. Benefiting from automated geolocation correction before reconstruction, Ozcanli et al. [13] acquired the elevation value corresponding to each pixel of satellite images by using a probabilistic volume reconstruction. In this method, the elevations are calculated by taking the median value point by point on multiview DSMs. However, the single acquisition mode is susceptible to noises in the multidate DSMs, producing incoherent DSM. Subsequently, Facciolo et al. [2] proposed a multimodality method to select the suitable modes corresponding to elevations based on adaptive k-medians clustering. The method is only able to fuse elevations at a single pixel location by adaptive clustering, which is hardly possible to solve the problem of elevation inconsistency between adjacent points. Consequently, Qin [14] proposed an adaptive median filtering method for resolving the elevation differences between adjacent points and preserving the object boundaries. This method first creates adaptive windows, and then, the median filter is applied to acquire elevation value within this window. Subsequently, the matching uncertainty in the stereo 3-D reconstruction stage was introduced into the method for selecting more reliable elevation values [15]. In this way, some of the outliers can be suppressed and a DSM with lower root-mean-square error (RMSE) is obtained by this modified method. Nevertheless, adaptive windows are prone to blurred edges and weak textures, resulting in edge distortion and loss of surface details.

To accurately locate region edges and preserve surface details, we adopt a coarse-to-fine clustering strategy, i.e., cluster the salient domain first, and then carry out model domain clustering within each salient region [29]. On the one hand, the salient regions can be utilized to locate object boundaries with dramatic elevation drop and provide edge constraint for fusion. On the other hand, the model domains with surface structures are able to be employed to provide model constraint and rebuild detail surface structures in weakly textured areas. Compared with the clustering-based approach proposed by Liu et al. [30], our method is independent of the shooting patterns, instead of only applicable to aerial oblique images and limited by the main optical axis of the shot.

In addition to the above two kinds of multiview 3-D reconstruction methods, deep learning-based multiview stereo (MVS) is also an effective method, due to the rapid development of deep learning in recent years [31], [32]. These learning-based reconstruction approaches distinguishably train an end-to-end mapping from images to depth maps or 3-D meshes. Nevertheless, deep learning-based MVS methods still lack generality in 3-D reconstruction based on satellite images due to the limitations in availability of training data about satellite images and corresponding ground-truth depth maps.

III. Methodology

The implementation of the proposed multiview DSM fusion method relies on the interaction of satellite images and DSMs; hence, the alignment serves as the first stage in the proposed methodology. In the second stage, the CDC, composed of salient domain clustering and model domain clustering, concurrently delineates significantly differing terrains by pinpointing distinct object edges and consolidates the detailed regions and their corresponding geometric models. In the concluding stage, the ultimate DSMs are obtained through multiview DSM fusion using WLS with the model clusters as fundamental units. An overview of the proposed algorithm is shown in Fig. 2.

A. Registration

First of all, the multiview DSMs need to be aligned with each satellite images, based on which the joint data of satellite imagery and DSM can be segmented. Establishing mapping relationships from object to satellite image space through rational function model (RFM) has become the de

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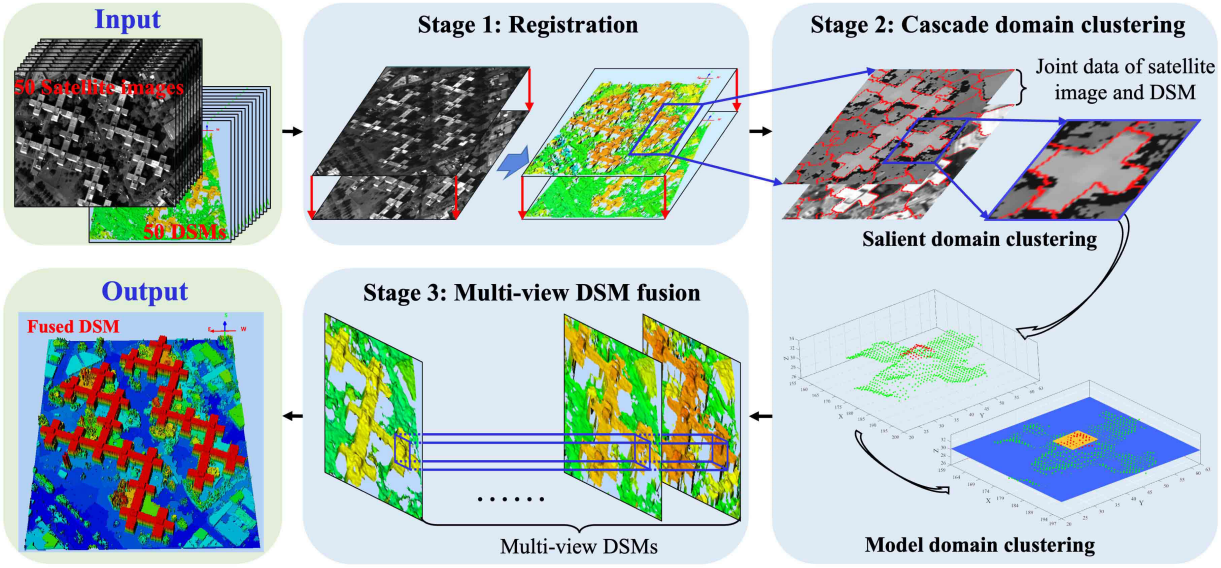


Fig. 2. Outline of our method.

facto standard, and satellite image suppliers are also able to provide the RPC used to construct the RFM. Referring to De Franchis et al. [33], let (B_n, L_n, H_n) be the normalized latitude, longitude, and elevation coordinates of a point

$$\begin{aligned} B_n &= (\varphi - B_\text{OFF})/B_\text{SCALE} \\ L_n &= (\lambda - L_\text{OFF})/L_\text{SCALE} \\ H_n &= (h - H_\text{OFF})/H_\text{SCALE} \end{aligned} \tag{1}$$

where φ, λ, and h are the latitude, longitude, and elevations, respectively. B_OFF, L_OFF, and H_OFF are the normalized offset parameters which stored in the satellite image profiles. B_SCALE, L_SCALE, and H_SCALE are the normalized scale parameters, which stored in the satellite image profiles, and then, the polynomial equation of RFM is

$$\begin{aligned} l &= \frac{L_{\text{num}}(B_n, L_n, H_n)}{L_{\text{den}}(B_n, L_n, H_n)} \\ s &= \frac{S_{\text{num}}(B_n, L_n, H_n)}{S_{\text{den}}(B_n, L_n, H_n)} \end{aligned} \tag{2}$$

where l and s are the image coordinates of the projection point, and $L_{\text{num}}(\cdot)$, $L_{\text{den}}(\cdot)$, $S_{\text{num}}(\cdot)$, and $S_{\text{den}}(\cdot)$ are the 20-D vectors holding the third-degree polynomial expansion of those normalized coordinates.

However, the DSMs obtained from different satellite image pairs are usually not aligned and vary in size due to pointing errors in the RFM. The affine transformation-based bias compensation is utilized to correct registration errors. Above all, geographical coordinates and elevation information stored in DSMs are first separated, and elevation values displayed as images by translating the elevation into grayscale. Then, the difference of grayscale distribution between different DSMs is reduced through histogram specification. After that the matching points between different DSMs are obtained via affine scale invariant feature transform (ASIFT) matching algorithm [34], as shown in Fig. 3.

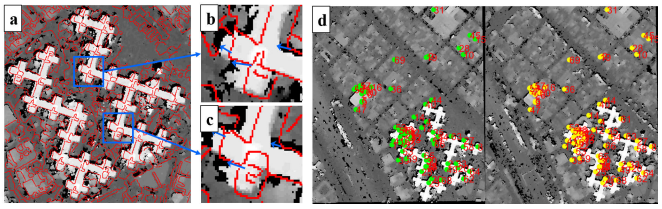


Fig. 3. Registration errors and ASIFT matching points. (a) To observe the registration errors between different DSMs, the canny edges are extracted and plotted on the other DSM. (b) and (c) Close-ups of the blue box in (a), and the blue arrows in (b) and (c) indicate the offsets. (d) ASIFT matching points in different DSMs (shown with ten times sparseness) are represented by dots with different colors, and the corresponding points are marked with the same numerical labels.

Finally, the first-order affine transformation is fit by the random sample consensus (RANSAC) [35]

$$\begin{bmatrix} l' \\ s' \\ 1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} l \\ s \\ 1 \end{bmatrix} \tag{3}$$

where A_{11}, A_{12}, A_{13}, A_{21}, A_{22}, and A_{23} are the coefficients of the affine transformation matrix, and (l', s') is the coordinate of a corrected point. Furthermore, the utilization of affine transformation to execute a rectification on the image has emerged as a widely adopted approach for image rectification. This method has been rigorously validated and demonstrated to possess a high degree of reliability, achieving a correction accuracy of 1/10 pixel [33]. Additionally, at least four pairs of nonlinear correspondence points between the two views are required to fit the affine fundamental matrix. Therefore, even in cases where local low building areas do not yield uniform ASIFT matching points [as depicted in Fig. 3(d)], a blend of dense and sparse matching points can still be utilized to fit affine fundamental matrix through RANSAC random search mechanism. Conversely, prior to registration, the maximum alignment error between DSMs of WV-3 satellite images with a GSD of 0.5 m is measured at 66.68 pixels, with an average

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error of 1.40 pixels. This measurement is derived from the calculation of the Euclidean distance between ASIFT match points on different DSMs.

B. Cascade Domain Clustering

The proposed CDC is formed by cascading complementary salient domain clustering and model domain clustering. The salient domain clustering is the first level of clustering, which achieves salient region delineation utilizing the spectral and 3-D location features of the points. Model domain clustering is the second level of clustering, which facilitates the aggregation and model fitting of points with the same surface structure by exploiting the underlying geometrical model of the points within salient regions.

1) Salient Domain Clustering: A widely known clustering-based segmentation algorithm is simple linear iterative clustering (SLIC) [36]. In most cases, the superpixels produced by SLIC are visually satisfying and the computational complexity is $O(n)$. Afterward, Chen et al. [37] proposed linear spectral clustering (LSC) superpixel algorithm, which connects local feature-based operation from SLIC to the global optimization objective function (OF); thus, the global image structure is able to be utilized to generate more reasonable segmentation results. Inspired by the two well-known algorithms mentioned above, salient domain clustering employs iterative weighted K-means clustering for feature aggregation, and guaranteed to have a time complexity of $O(n)$.

The original LSC algorithm involves the transformation of standard image pixels into a 10-D feature space, followed by the implementation of weighted K-means for segmentation. This ensures a theoretical equivalence between the weighted K-means clustering in the 10-D feature space and the normalized cuts in the original pixel space, thereby enabling the preservation of nonlocal information [37]. Consequently, the weighted K-means clustering can optimize the cost function as defined by the normalized cut formulation.

To effectively cluster salient regions in DSM without compromising the global structure, the proposed salient region clustering method initially involves the mapping of each pixel in the joint data of satellite imagery and DSM into a 12-D feature pace. The points that exhibit proximity to each other in the 12-D feature space are subsequently clustered through the utilization of weighted K-means clustering. With the utilization of spectral information from satellite images, the salient domain clustering method is capable of delineating distinct terrain areas, even in cases where stereo DSMs are affected by incomplete representation and missing values.

In detail, the point p is first denoted by a 5-D vector. To obtain accurate salient regions in line with actual terrains, we add an additional elevation feature e_p based on the LSC original $(l_p, \alpha_p, \beta_p, x_p, y_p)$, i.e., $p = (l_p, \alpha_p, \beta_p, x_p, y_p, e_p)$, where l, α, and β are the color components in CIELAB color space. x and y are the horizontal and vertical coordinates in image space, respectively. A 12-D feature space can be reached by the new mapping function ϕ_e, as follows:

$$\phi_e(p) = \frac{1}{\omega(p)} \begin{bmatrix} C_c \cos\frac{\pi}{2} l_p, C_c \sin\frac{\pi}{2} l_p, \\ \times 2.55 C_c \cos\frac{\pi}{2}\alpha_p, 2.55 C_c \sin\frac{\pi}{2}\alpha_p, \\ \times 2.55 C_c \cos\frac{\pi}{2}\beta_p, 2.55 C_c \sin\frac{\pi}{2}\beta_p, \\ \times C_s \cos\frac{\pi}{2} x_p, C_s \sin\frac{\pi}{2} x_p, \\ \times C_s \cos\frac{\pi}{2} y_p, C_s \sin\frac{\pi}{2} y_p, \\ \times C_e \cos\frac{\pi}{2} e_p, C_e \sin\frac{\pi}{2} e_p \end{bmatrix} \tag{4}$$

where C_e is utilized to control the importance of elevation against color information. In practice, ratio $r_e = C_e/C_c$ is employed to balance the weight between elevation and color information. In general, the larger the r_e is, the more pixels with similar elevation tend to be clustered into one class. The best value of r_e is acquired through repeated experiments. To make parameter r_e robust to DSMs with different altitudes and satellite images with different exposures, the range of elevation values in DSM is first linearly normalized to $[0, 1]$, and then projected into the color space of the satellite image (e.g., $[0, 2^8 - 1]$ or $[0, 2^{16} - 1]$) by means of grayscale histogram stretching. C_c and C_s are used to control the relative significance between color and spatial distance. In practice, the ratio $r = C_c/C_s$ is utilized to balance the weights of color and spatial information. The larger the r is, the easier it is for spatially close pixels to be clustered into one class, yielding regions with more regular shapes. The best value of r is received through repeated experiments.

The weighted K-means algorithm is an expansion of the conventional K-means clustering approach, wherein each point p is assigned a weight $\omega(p)$ to denote its significance or reliability. Receiving inspiration from Aradnia et al. [38], in the salient domain clustering, the computation of the weights involves the distance between the data points and the clustering center, with the weights being inversely related to the distance. The detailed calculation of the weights is as follows:

$$\omega(p) = \begin{cases} 0, & \text{if } D_p = 0 \\ \dfrac{1}{\sum_{t=1}^{h} \frac{D_p}{D_t}}, & \text{if } D_p \neq 0 \end{cases} \tag{5}$$

where

$$D_p = \sum_{l=1}^{k} \sum_{i=1}^{n} u_{i,l} d(x_{i,p}, z_{l,p}) \tag{6}$$

where

$$0 \le u_{i,l} \le 1, \quad \sum_{l=1}^{k} u_{i,l} = 1, \quad 0 \le l \le k, \quad 1 \le i \le n \tag{7}$$

where h is the number of $D_p \neq 0$, k is the number of clusters, n is the number of variables in the cluster, and $d(\cdot)$ is the Euclidean distance function. For D_t, t denotes the variable from 1 to h. The primary purpose of D_t is to compute the cumulative sum $\sum_{t=1}^{h} (D_p/D_t)$.

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Subsequently, the K seed pixels denote the search centers are uniformly sampled from the entire image with horizontal and vertical intervals of v_x and v_y, respectively, where v_x/v_y approximately matches the image aspect ratio. The decision to limit the search space for salient domain clustering to a local region, rather than the entire image, is based on the requirement of local compactness for the presence of salient domains. Consequently, the search space for each salient domain is confined to $\tau v_x \times \tau v_y$, where $\tau > 0.5$ means a parameter that strikes a balance between local compactness and global optimality. Moreover, in conventional superpixel segmentation algorithms, the number of superpixel K exhibits a positive correlation with the accuracy of superpixel edges. In other words, a larger number of superpixels K corresponds to greater superpixel edge accuracy, provided that the image size remains constant. Nevertheless, upon surpassing an optimal value, the superpixel edge accuracy stabilizes, leading to the following relationship:

$$M \cdot N = K \cdot S^2 \tag{8}$$

where M and N represent the image size, and S represents the average size of the superpixel. In accordance with the inherent characteristics of the superpixels, a larger superpixel size increases the likelihood of more pixels within the superpixel deviating from the actual target. Hence, the primary factor influencing the accuracy of the superpixel is the average size of the superpixel, denoted as S. More importantly, the value of S is independent of the image size. In this manner, regardless of variations in $M \cdot N$, once the optimal S_{opt} is established, the adaptive optimal value of K can be calculated as follows:

$$K = M \cdot N / S_{\text{opt}}^2. \tag{9}$$

Subsequently, the optimal average size of superpixels (called salient domains in our method) S_{opt} will be determined through repetitive experiments conducted on images from various regions in Section IV-A2).

Following this, the Euclidean distance is employed to calculate the distance between each pixel and the clustering center in the next weighted K-means clustering, as follows:

$$F_{km} = \sum_{k=1}^{K} \sum_{p \in \pi_k} \omega(p) \left\| \phi_e(p) - c_k \right\|^2 \tag{10}$$

where each point is assigned a weight $\omega(p)$, K denotes the number of clusters, π_k denotes the kth ($k = 1, 2, \ldots, K$) cluster, and c_k is the center of the π_k, as follows:

$$c_k = \sum_{q \in \pi_k} \omega(q) \phi_e(q) \Big/ \sum_{q \in \pi_k} \omega(q). \tag{11}$$

Next, each pixel is assigned to the cluster that exhibits the closest proximity of its weighted mean to the pixel's vector in the 12-D feature space. Upon pixel assignment, the weighted mean and centroid of each cluster are adjusted accordingly. These two steps are iteratively performed until convergence is attained.

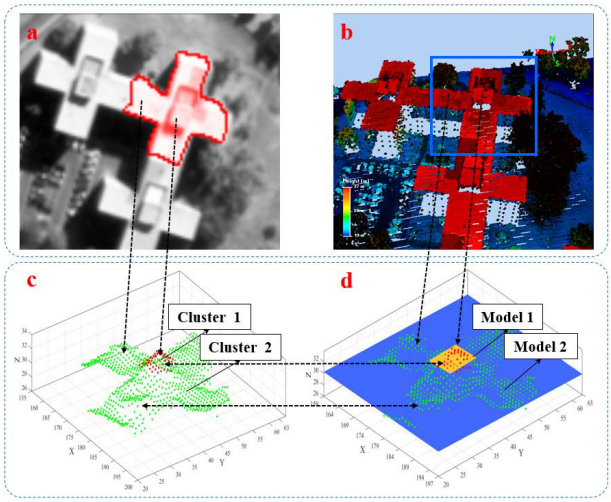


Fig. 4. Model domain clustering schematic. (a) Satellite imagery. (b) Ground-truth LiDAR point cloud. (c) Results of model domain clustering with different colors indicating different classes. (d) Results of model fitting. The point clouds in (c) and (d) correspond to the red area in (a) and the blue box in (b). The black dashed arrows indicate the corresponding class regions.

2) Model Domain Clustering: Clearly, salient domain clustering is unable to aggregate all ground object regions, especially fine objects with weak textures. Therefore, we transform the segmentation of detailed surfaces into a clustering of 3-D points with the same geometric structure (i.e., model domain clustering), and the clusters of 3-D pixels obtained by clustering are used as basic units for multiview DSM fusion. Moreover, with the aggregation effect of salient domains to similar pixels, the accuracy and speed of model domain clustering within salient regions are both assured. The model domain clustering schematic is shown in Fig. 4.

a) Clustering process: Due to the variability of cluster numbers within salient domains, traditional clustering algorithms that depend on a predetermined number of clusters are not well-suited to address this issue. Drawing inspiration from T-Linkage, the problem of clustering within the model domain can be transformed into a multiple instance model fitting problem, eliminating the requirement of specifying the number of models [29]. Following the T-Linkage clustering process, the minimum sample set (MSS) of points is first generated by random sampling. The m model hypotheses are as follows:

$$H = \{h_j\}_{j=1,\ldots,m}. \tag{12}$$

The estimated model hypotheses can be used as a hypothesis pool, and the consensus set (CS) of each model is computed within the hypothesis pool. Essentially, CS is the set of points whose distance from the model is less than a threshold $\varepsilon < R$, which allows the preference set (PS) of each point x to be calculated within the CS. The PS is a binary function, which indicates how much each point likes the model, like is 1 and dislike is 0. Furthermore, according to T-Linkage [29], the binary PS is capable of being generalized by extending it to the preference function (PF). Such that, each point x is described by PF, and its value is held in the complete closed interval [0, 1], with 1 being the most preferred and 0 being the least

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preferred. The PF is defined as follows:

$$\phi(h) = \begin{cases} e^{-d(x,h)/\tau}, & d(x,h) < 5\tau \\ 0, & d(x,h) \geq 5\tau \end{cases} \tag{13}$$

where τ represents the same function as threshold ε, and $d(x, h)$ denotes the residual between point x and model assumption h (given in detail in Section III-B2b). In this way, the point preferences can be described more accurately by calculating the residuals between PFs.

Starting with single instance x, model domain clustering carries out clustering in a bottom-up way. If the subset of points $S = \{x\}$, then the PF of S is equal to the PF of x denoted by q_x. Otherwise, the PF of S is the vector $p \in [0, 1]^m$, such that its ith component is

$$p_i = \min_{x \in S} (q_x)_i. \tag{14}$$

Subsequently, the similarity between the PFs $[0, 1]^m$ of each point is calculated by Tanimoto distance

$$d_T(p, q) = 1 - \frac{\langle p, q \rangle}{\|p\|^2 + \|q\|^2 - \langle p, q \rangle'} \tag{15}$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product of $\mathbb{R}^m$, and Tanimoto distance range is [0, 1]. For $\forall p, q \neq 0 \in [0, 1]^m$, if $p = q$ then $d_T(p, q) = 0$, if $p \perp q$ then $d_T(p, q) = 1$. Consequently, as long as there exists nonorthogonal p and q, all subsets of points will be scanned and the most similar p and q will be calculated by minimizing Tanimoto distance, as follows:

$$d_T(p, q) = \min_{r,s} d_T(r, s). \tag{16}$$

In the end, the most similar p and q are replaced with their union, and the PF and surface model of the new cluster are recomputed.

b) Geometric model: In the proposed geometric model, the specific meaning of the model assumption h in 13 is the 3-D geometric model, and the geometric model $ax+by+cz+d = y$ in 3-D space can be achieved through singular value decomposition (SVD).

Initially, the centroid $(\bar{x}, \bar{y}, \bar{z})$ of the 3-D point set $\{(x_i, y_i, z_i)\}_{i=1,\ldots,n} (n \geq 3)$ is calculated. Next, the vector $N = [a, b, c]^{\mathrm{T}}$, consisting of the coefficients a, b, and c, represents the normal vector of the geometric model, as dictated by the inherent properties of the model. By considering the characteristics of normal vector, we can derive the following equation:

$$\begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} & z_1 - \bar{z} \\ x_2 - \bar{x} & y_2 - \bar{y} & z_2 - \bar{z} \\ & \cdots & \\ x_n - \bar{x} & y_n - \bar{y} & z_n - \bar{z} \end{bmatrix} \begin{bmatrix} a \\ b \\ C \end{bmatrix} = 0. \tag{17}$$

Let the left matrix be A, the OF can be expressed as $\min \|AN\|$. Due to $\|N\| = 1$, the SVD of A can be solved to achieve $A = UDV^{\mathrm{T}}$, where D is the diagonal array, and U and V are both unitary matrices. According to the properties of unitary matrix, we can get

$$\|AN\| = \left\|UDV^{\mathrm{T}}N\right\| = \left\|DV^{\mathrm{T}}N\right\| \tag{18}$$

when $\|V^{\mathrm{T}}N\| = \|N\| = 1$ and $V^{\mathrm{T}}N = [0\ 0\ 1]^{\mathrm{T}}$, $\|AN\|$ can take the minimum value. At this time, the diagonal elements of D are singular values, the last diagonal element is the minimum singular value, and thus, the singular vector corresponding to the minimum singular value is the normal direction of the plane. Eventually, the coefficients a, b, and c can be yielded as follows:

$$N = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = V \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{19}$$

then the constant term $d = -[\bar{x}, \bar{y}, \bar{z}]N$ can be achieved.

In the task of model domain clustering, the residual formula in (13) is specifically meant to be the distance between points to the plane. And since $\|N\|_2 = (a^2 + b^2 + c^2)^{1/2} = 1$, then the formula for distance is as follows:

$$d(x, h) = |ax_i + by_i + cz_i + d|. \tag{20}$$

In conclusion, the process of the CDC is summarized in Algorithm 1.

C. Multiview DSM Fusion

The outcome of CDC reveals that each model cluster corresponds to a distinct geometric structure. Conversely, a geometric structure may span across multiple model clusters, as the division of a complete structure into several model clusters is a consequence of CDC. Hence, it is adequate and unnecessary for 3-D points within a single model cluster to exhibit consistency in both geometric structure and surface texture. This property enables the model cluster to function as the basic unit for DSM fusion. Consequently, WLS is employed to obtain precise elevations while considering the constraint of a single class region, which is calculated as follows.

Initially, the initial DSM D_{ini} is derived through the application of median filtering, while outliers are eliminated by considering the distance between points and the geometric model. Subsequently, the bilateral filtering technique is applied as the OF to effectively reduce noise and obtain accurate elevations. Following this, the RF is utilized to estimate the deviations between the objective elevations and the stereo DSM elevations, enabling an iterative approximation based on the objective elevations in the opposite directions of the deviations. This iterative process exemplifies the concept of iteratively solving for the minimum of the RF along the directions opposite to the residuals, as shown in Fig. 5.

Specifically, the bilateral filter works as an OF to estimate the objective elevations, as follows:

$$\mathrm{OF}[e]_p = \hat{e}_p^{\mathrm{bf}} = \frac{1}{W_p^{\mathrm{bf}}} \sum_{q \in \mathcal{S}} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}\left(\left|e_p - e_q\right|\right) e_q \tag{21}$$

where e_p and e_q are the elevations of points p and q, respectively. $\mathcal{S}$ denotes the neighborhood centered on point p, and W_p^{bf} denotes the normalized weight, as follows:

$$W_p^{\mathrm{bf}} = \sum_{q \in \mathcal{S}} G_{\sigma_s}(\|p - q\|) G_{\sigma_r}\left(\left|e_p - e_q\right|\right) \tag{22}$$

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Algorithm 1 CDC

Input: Satellite image I, DSM D
Output: Label set of model clusters $\mathbb{M}$, geometric model set $\mathbb{G}$.

```
// The salient domian clustering based on LSC
```

Map each point $p = (l_p, \alpha_p, \beta_p, x_p, y_p, e_p)$ to 12D vector $\phi_e(p)$ by Eq. 4 and 5;
Obtain the DSM dimensions M and N and calculate K according to Eq. 9;
Sample K seed points uniformly on the whole image I and D with horizontal and vertical intervals of v_x and v_y;
Initialize distance $d(p) = \infty$ for each point p and each cluster center c_k by Eq. 11;
do
  for *each point p in K seed regions* $\tau v_x \times \tau v_y$ **do**
    Caculate euclidean distance $d_e(p)$ between $\phi_e(p)$ and c_k by Eq. 10;
    if $d_e(p) < d(p)$ **then**
      Update distance $d(p) = d_e(p)$ and label $L(p) = k$;
  Update all cluster center c_k by Eq. 11;
while *the K-means dose not converge*;

```
// The model domain clustering
```

for *each salient region* $\mathcal{R}_k$ **do**
  Get 3-D point set $\mathbb{P}_k = \{(x, y, z)|\mathrm{L}(x, y, z) = k\}$ in region $\mathcal{R}_k$;
  Initialize each point p, $p \in \mathbb{P}_k$ in its own cluster and compute their PF by Eq. 13 ;
  do
    Find p, q by Eq. 15 and 16;
    Replace these two clusters with their union and update the PF, label $l_m, l_m \in \mathbb{M}$, geometric model $m_g, m_g \in \mathbb{G}$ of this new cluster;
  while $\exists$ *PF not orthogonal*;
  Outliers rejection;
return $\mathbb{M}$ and $\mathbb{G}$;

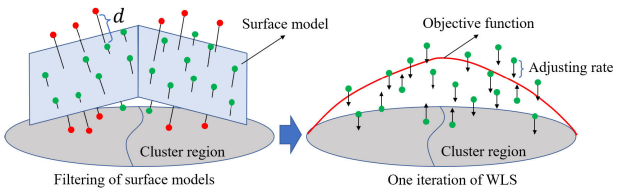


Fig. 5. Multiview DSM fusion process. (Left) Two clusters obtained by surface model clustering, accompanied by its corresponding surface model. Inline points (green points) and outliers (red points) are distinguished via the distance. (Right) After the OFs and deviation directions are calculated, the 3-D points within the class regions are iteratively optimized.

where $G_{\sigma_s}(\cdot)$ and $G_{\sigma_r}(\cdot)$ representthe spatial domain and the elevation range domain, respectively, as follows:

$$G_{\sigma_s}(\|p - q\|) = \exp\left(-\frac{(x-u)^2 + (y-v)^2}{2\sigma_s^2}\right) \tag{23}$$

$$G_{\sigma_r}\left(\left|e_p - e_q\right|\right) = \exp\left(-\frac{\left(e_p - e_q\right)^2}{2\sigma_r^2}\right) \tag{24}$$

where σ_s and σ_r denote the standard deviation of the spatial domain and the elevation range domain, respectively. The coordinates of points p and q are (x, y) and (u, v), respectively.

The bilateral filtering results combine the dual effects of the spatial domain and the elevation range domain, i.e., in flat areas, elevations change slightly, and the weight of corresponding elevation range domain are close to 1. Thus, the weight of the spatial domain plays a major role approximating Gaussian filtering. At the edges of the buildings, the elevations vary greatly, and the weight of elevation range domain become larger, thus maintaining the information of the edges.

After bilateral filtering, the directions of the deviations ∇ between the objective elevations and the stereo DSM elevations are calculated as follows:

$$\nabla = \frac{\text{res_val}}{|\text{res_val}|} \tag{25}$$

where res_val denotes the degree of deviation, which is calculated by the RF $\mathrm{RF}(\cdot)$ as follows:

$$\text{res_val} = \mathrm{RF}\left(e_j, \mathrm{OF}[e]\right) = \sum_{j}^{N} \omega_g \cdot \left(e_j - \mathrm{OF}[e]\right). \tag{26}$$

The RF $\mathrm{RF}(\cdot)$ is defined by the sum of the deviations between the elevations from stereo DSMs and the OFs $\mathrm{OF}(\cdot)$. To mitigate the influence of outlier 3-D points on the deviation direction, we employ a strategy during the accumulation of deviation values that assigns greater weights to 3-D points in close proximity to the true elevation, while assigning smaller weights to those further away. This is achieved by regulating the weight of the difference using the Gaussian function, considering that elevations from different stereo DSMs conform to a Gaussian distribution at a specific location. The computation of Gaussian weights is executed as follows:

$$\omega_g = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left(-\frac{(e-\mu)^2}{2\sigma^2}\right) \tag{27}$$

where σ denotes the standard deviation of the elevations and μ denotes the mean of the elevations.

Consequently, the objective elevation can be corrected in the reverse direction of ∇, aiming to minimize the disparity between the objective elevations and the true elevations, as illustrated in the following:

$$e_i = -\text{rate} \cdot \nabla_i + e_{i-1} \tag{28}$$

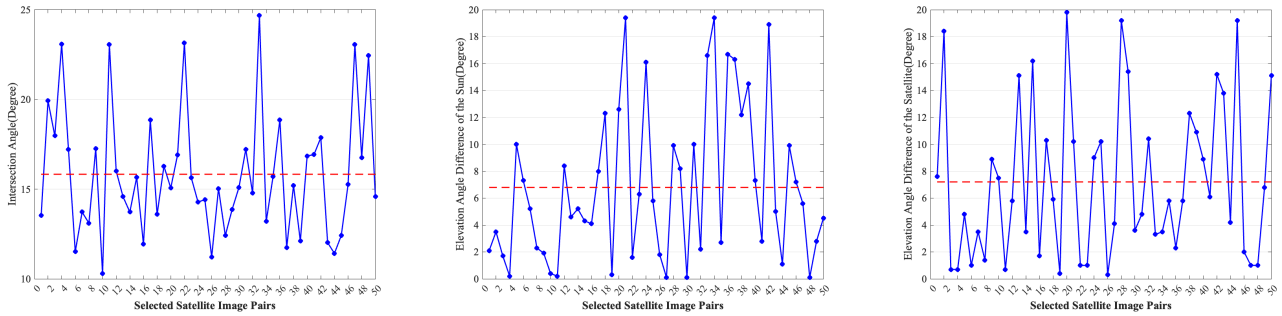
where e_i and e_{i-1} are the elevations of the ith$(i \geq 2)$ and $(i-1)$th iterations, respectively, rate denotes the adjusting rate of each iteration, and ∇_i denotes the direction of the difference between the objective elevation and the stereo DSM elevation in the ith iteration.

In summary, the multiview DSM fusion based on CDC is summarized in Algorithm 2.

IV. Experiments

In this section, the proposed method is evaluated qualitatively and quantitatively based on WV-2 and WV-3 satellite images provided by the Johns Hopkins University Applied Physics Laboratory (JHUAPL), Laurel, MD, USA [39]. WV-2

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(a) (b) (c)

Fig. 6. Intersection angles, elevation angle differences of the Sun, and elevation angle differences of the satellite for 50 pairs of satellite images. The red dashed lines indicate the mean values. (a)–(c) Statistics for intersection angles, elevation angle differences of the Sun and elevation angle differences of the satellite.

Algorithm 2 Multi-View DSM Fusion

Input: Multi-view DSMs $\{D_n\}_{n=1,\ldots,N}$, label set of model clusters $\mathbb{M}$, geometric model set $\mathbb{G}$, threshold ε, the maximum iteration number itr_{max}

Output: Fused DSM D_f

for *each model cluster* $l_m, l_m \in \mathbb{M}$ *and geometric model* $m_g, m_g \in \mathbb{G}$ **do**
  Compute initial DSM D_{ini} by median filtering and outlier filtering based on m_g;
  repeat
    for *each pixel* p *inside model cluster* l_m **do**
      Estimate the objective elevations $\mathrm{OF}[e]_p$ based on D_{ini} by Eq. 21;
      Calculate directions of the deviations ∇ by Eq. 25, 26 and 27;
    Update the elevation e by Eq. 28;
  until *the degree of deviation* $res_val < \varepsilon$ *or reach the maximum iteration number* itr_{max};
return D_f;

comprises three panchromatic images, covering an area of approximately 64 km^2 in Washington, DC, USA. However, this dataset lacks authentic data for quantitative assessment. WV-3 includes 50 panchromatic and multispectral digital global images, which covers approximately 100 km^2 near San Fernando, Argentina. These satellite images with significant seasonal differences were acquired by both in-orbit and cross-orbit photography, spanning from November 2014 to January 2016. To enable quantitative assessment of the accuracy and completeness metrics of the reconstruction algorithms, the dataset also encompasses ground-truth airborne LiDAR point clouds with the nominal point spacing of 20 cm and the coverage area of 20 km^2.

Considering that the multiview DSM fusion are based on stereo 3-D reconstruction of satellite imagery, 50 pairs of satellite images from a combination of 43 different viewpoints are selected for obtaining stereo DSMs. Moreover, the intersection angle, elevation angle difference of the Sun, and elevation angle difference of the satellite corresponding to each pair of images are shown in Fig. 6. Obviously, the values of the intersection angle, elevation angle difference of the Sun, and elevation angle difference of the satellite lie in the interval $[10°, 25°]$, $[0°, 20°]$, and $[0°, 20°]$, respectively. For that in densely built environments, relatively smaller intersection angles, solar and satellite inclination differences typically ensure image pairs exhibiting smaller differences, consequently yielding better DSMs in dense reconstruction [40]. Nevertheless, too diminutive intersection angles to furnish an ample amount of stereo information for high-rise buildings, thereby influencing the precision of stereo 3-D reconstruction.

In addition, the satellite stereo pipeline (S2P) in [41] is utilized for stereo 3-D reconstruction to generate stereo DSMs for multiview DSM fusion. Afterward, the stereo 3-D reconstruction results are utilized as the input of the proposed algorithm. The experiments in this article are run on a personal computer equipped with an Intel Core i7-8700 (3.20 GHz) CPU and a 16G RAM. We do not use any parallel software, GPUs, or other specialized devices.

A. Verification Experiments on Parameters as Well as Sensitivities

Prior to conducting the evaluation experiments on DSM fusion, experimental validation is essential for determining the optimal values of the salient domain clustering parameter r and r_e, the salient region number K, the number of DSMs, and the number of iterations for WLS $\mathrm{itr}_{\max}$.

1) Determination of Parameters r and r_e: According to Section III-B1, the parameters r and r_e are employed to achieve a balance between the weights assigned to color and spatial information, as well as the weight assigned to elevation and color. Therefore, the optimal value of parameter r is first obtained by controlling $r_e = 0.5$ representative results that are selected from numerous repeated experiments, as shown in Fig. 7.

Based on Fig. 7, it can be observed that increasing the value of parameter r results in smoother edges for salient regions; however, this comes at the cost of reduced accuracy. To strike a balance between edge smoothness and accuracy, we set the value of parameter r to 0.4. Subsequently, we conduct additional iterations of the salient domain clustering experiments with the condition $r = 0.4$, applying various values of parameter r_e. From the multitude of repeated experiments,

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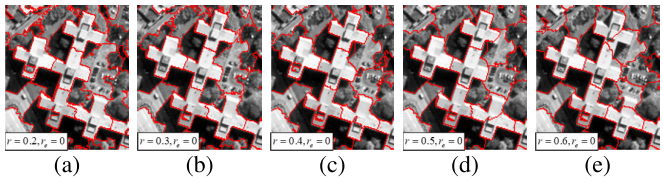


Fig. 7. Results of salient domain clustering when $r_e = 0$ and r changes incrementally. To assess the precision of the salient domain edges, the edges are visualized as red lines and projected onto the satellite image. (a)–(e) Results of salient domain clustering with r values of 0.2, 0.3, 0.4, 0.5, and 0.6, respectively.

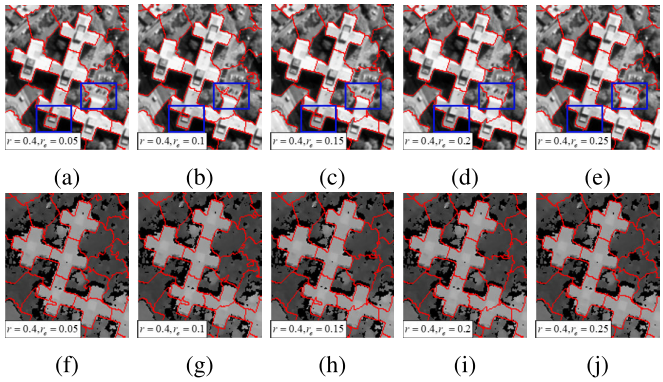


Fig. 8. Results of salient domain clustering when $r = 0.4$ and r_e changes incrementally. The first row shows the edges of salient regions projected onto the satellite image, and the second row shows the edges of salient regions projected onto the DSM. The blue boxes marked the partially changed edges. (a)–(e) Results on satellite images with re values of 0.05, 0.1, 0.15, 0.2, and 0.25, respectively. (f)–(j) Results on DSMs with re values of 0.05, 0.1, 0.15, 0.2, and 0.25, respectively.

we select five representative results, which are depicted in Fig. 8.

As shown in Fig. 8, only the smoothness of the edges of buildings goes up as the parameter r_e increases (e.g., blue box areas of Fig. 8). It is explained by the fact that only the elevations of building edges changed drastically. Moreover, the edges are most accurate when the value of r_e is 0.2, thus the parameters of salient domain clustering are set to $r = 0.4$ and $r_e = 0.2$.

Furthermore, the parameters r and r_e exhibit resilience to regional variations. We have considered the variations in DSM values resulting from changes in regions, and therefore, the stereo DSMs are preprocessed to consistently maintains the same scale as the satellite image ($[0, 2^8 - 1]$ or $[0, 2^{16} - 1]$) by means of scale transformation, as outlined in Section III-B. In this way, the DSMs used in salient domain clustering are no longer absolute elevation values, but rather relative elevation values. Despite regional changes affecting elevation, the relative elevation domain remains relatively constant with the color domain. Consequently, the parameters r and r_e, which are utilized to balance the ratio of elevation to color values, remain unaffected by regional variations, as confirmed through repetitive experiments.

2) Determination of the Optimal Average Size of Salient Domains S_{opt}: In order to quantitatively evaluate the impact of S on edge accuracy, four metrics are employed. These metrics, namely, under-segmentation error (UE) [36], boundary recall (BR) [42], achievable segmentation accuracy (ASA) [42], and intersection-over-union (IoU) [43], are utilized to assess the quality of the segmented edges. The calculation for each metric is as follows:

$$\mathrm{UE} = \frac{\left[\sum_{\{s_j | s_j \cap g_i \neq \emptyset\}} \mathrm{Area}(s_j)\right] - \mathrm{Area}(g_i)}{\mathrm{Area}(g_i)} \tag{29}$$

where $g_1, \ldots, g_K$ represent the labels of ground-truth data, and $s_1, \ldots, s_L$ represent the labels of regions. The UE evaluates the segmentation quality based on the requirement that the region overlap with only one object, with a lower UE indicating fewer regions spanning multiple objects

$$\mathrm{BR} = \frac{\sum_{g \in B_G} \mathrm{IF}(\min \|s - g\| < \varepsilon)_{s \in B_S}}{|B_G|} \tag{30}$$

where B_G and B_S denote the edges of ground truth and salient regions, respectively. g and s denote the elements of the sets B_G and B_S, respectively. $\mathrm{IF}(\cdot)$ is a binary function, if the distance between the region edge and the ground-truth edge is less than $\varepsilon (\varepsilon = 2)$, then $\mathrm{IF}(\cdot) = 1$; otherwise, $\mathrm{IF}(\cdot) = 0$. As a result, BR has the ability to measure the recovery of superpixel edges from ground-truth edges

$$\mathrm{ASA} = \frac{\sum_i \mathrm{Area}(G_j \to \max_i(S_i \wedge G_j))}{\sum_j \mathrm{Area}(G_j)} \tag{31}$$

where S_i and G_j represent the label of the ith region and the jth ground truth, respectively. $G_j \to \max_i(S_i \wedge G_j)$ denotes that the ith salient region which has the largest overlap with the jth ground truth is marked with the label of the jth ground truth. Hence, ASA can effectively measure the precision of the salient region by determining the ratio between the maximum overlapping salient region and the ground-truth regions

$$\mathrm{IoU} = \frac{|G \cap S|}{|G \cup S|}. \tag{32}$$

Here, G represents one of the actual object regions, and S denotes the set of all salient regions that intersect with G. It is important to note that salient domain clustering does not entail object-specific segmentation. Therefore, the average of multicategory IoU (mIoU) is employed to assess the accuracy of region contours across the entire image as follows:

$$\mathrm{mIoU} = \frac{1}{N} \sum_i^N \mathrm{IoU}_i \tag{33}$$

where N denotes the number of actual objects.

Moreover, it is true that publicly available datasets for evaluating segmentation accuracy are primarily designed for ordinary images and may not be suitable for joint data analysis of satellite imagery and DSM. To address this limitation, we have created a standard dataset[1] that includes ground-truth segmentation results of WV-3 satellite images along with their corresponding DSMs. This dataset can be used to evaluate the segmentation performance specifically for joint satellite imagery and DSM analysis. To ensure objectivity in manual segmentation, each region within the dataset is segmented by five different individuals, resulting in five distinct ground-truth results. The satellite images within the dataset encompass not only artificial structures like buildings and roads but also various vegetation. By averaging these

[1] https://github.com/zhao1i/Superpixel-dataset.git

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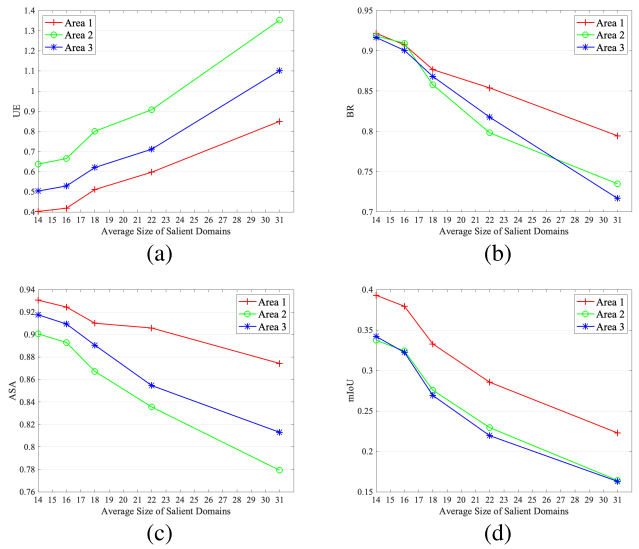


Fig. 9. Quantitative evaluation results of the salient domain clustering under the conditions of average size of salient domains from 14 to 31 pixels. (a)–(d) Curves of UE, BR, ASA, and mIoU in sequence.

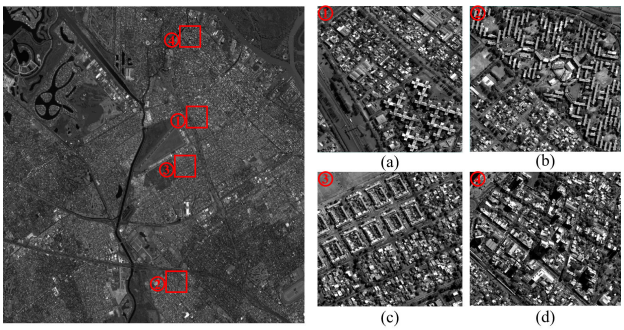


Fig. 10. Distribution of test areas on the WV-3 satellite image. (a)–(d) Areas 1–4 in sequence. Areas 1 and 2 are distinguished by densely clustered mid-rise buildings, while Area 3 is marked by densely distributed low-rise buildings. Area 4, on the other hand, is characterized by sparsely scattered high-rise buildings.

evaluations, a more objective assessment is achieved. Additionally, following the quantitative evaluation approach used for the superpixel algorithm, we assessed the accuracy of clustered salient domains at average size from 14 to 31 pixels, as illustrated in Fig. 9.

Based on the observations depicted in Fig. 9, the relationship between the size of the salient domains and the edge accuracy of the aggregated salient domains is inversely proportional. Intuitively, when each salient domain contains only one pixel, the edge accuracy is maximized. However, this extreme scenario is impractical as it leads to excessively high computational complexity and a greater risk of cutting off the targets, thereby affecting the subsequent clustering of the model domain. Upon analysis of Fig. 9, it is apparent that the rate of change in salient domains accuracy decreases when the size $S < 16$. $S = 16$ represents a pivotal inflection point for the four subplots of Fig. 9. The validity of this inflection point will be further examined across four distinct areas in Fig. 10, with sizes of 599 × 683, 614 × 631, 451 × 446, and 681 × 617, respectively.

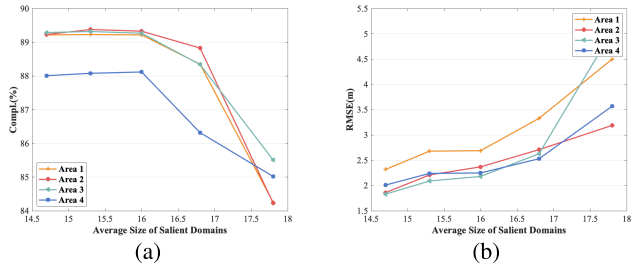


Fig. 11. Variation curves of Compl. and RMSE with the average size of salient domains. (a) and (b) Line graph of Compl. and RMSE, respectively.

TABLE I
PARAMETERS FOR TRIALS VARYING THE SIZE OF SALIENT DOMAINS

Parameters	Description	Value
r	Parameter of balancing the weight between color and spatial information	0.4
r_e	Parameter of balancing the weight between elevation and color information	0.2
ε	Threshold for iteration termination in WLS	0.5m
S	Average size of salient domains	**Variable**
itr_{max}	Maximum number of iterations in WLS	50

During the sensitivity experiments, we assigned multiple values to S to observe the variations in DSM accuracy. Additionally, the evaluation metrics completeness (Compl.) proposed by Bosch et al. [39] and RMSE are used to assess the fused DSMs with different parameters, and the calculations are given as follows.

1) Compl. is achieved by calculating the percentage of the number of nonempty cell errors within the threshold δ

$$\text{Compl.} = \frac{1}{N}\sum_{i=1}^{N} \text{IF}(|E_i - \text{Truth}_i| < \delta) \times 100\%. \quad (34)$$

2) RMSE is the mean value of the square of all elevation errors

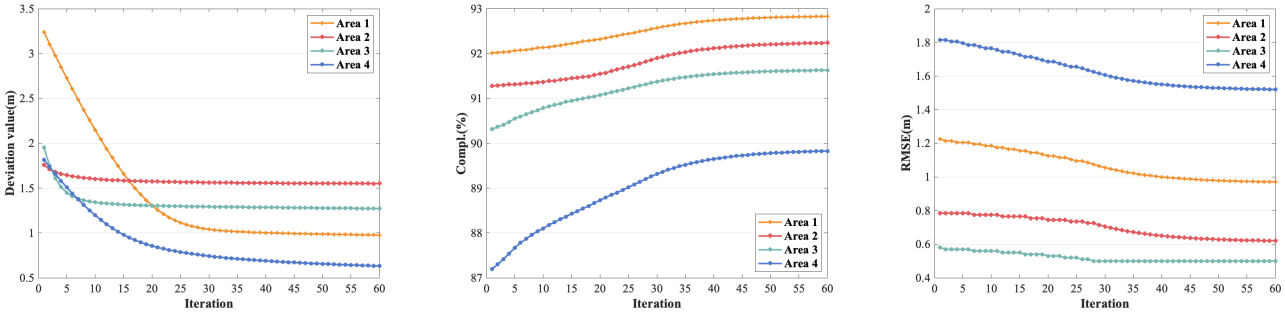
$$\text{RMSE} = \sqrt{\frac{1}{N}\sum (E_i - \text{Truth}_i)^2} \quad (35)$$

where E_i and Truth_i denote the elevation value and the ground-truth value at the ith nonempty cell, respectively. N is the number of nonempty cells. IF$(\cdot)$ is a binary function; if $|E_i - \text{Truth}_i| < \delta(\delta = 3)$, then IF$(\cdot) = 1$; otherwise, IF$(\cdot) = 0$.

Furthermore, for WV-3 images, the LiDAR point clouds can serve as ground-truth reference measurements for inclusion in accuracy calculation. Moreover, alignment of the point clouds with the DSM is necessary prior to evaluation, as the resolution of the LiDAR point cloud is considerably finer than that of the DSM. First, the coordinates of multiple 3-D points in point cloud are combined into a single matrix cell from an empty matrix which has the same size as the DSM. Then, the value of each matrix cell is set to the median Z value of all 3-D points within that cell. If the matrix cell is empty, the cell will be marked as NaN. The variation curves of Compl. and RMSE of fused DSM are shown in Fig. 11, and the remaining parameters utilized are delineated in Table I.

Based on Fig. 11, it is evident that the completeness of the fused DSM decreases while the RMSE increases as the average

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(a) (b) (c)

Fig. 12. Curve diagrams of deviation value, Compl., and RMSE corresponding to different numbers of iterations. (a)–(c) Changes of deviation value, Compl., and RMSE, respectively.

TABLE II
PARAMETERS FOR TRIALS VARYING THE NUMBER OF ITERATIONS

Parameters	Description	Value
r	Parameter of balancing the weight between color and spatial information	0.4
r_e	Parameter of balancing the weight between elevation and color information	0.2
ε	Threshold for iteration termination in WLS	0.5m
S_{opt}	The optimal average size of salient domains	16
itr_{max}	Maximum number of iterations in WLS	**Variable**

TABLE III
PARAMETER VALUES FOR MULTIVIEW DSM FUSION

Parameters	Description	Value
r	Parameter of balancing the weight between color and spatial information	0.4
r_e	Parameter of balancing the weight between elevation and color information	0.2
ε	Threshold for iteration termination in WLS	0.5m
S_{opt}	The optimal average size of salient domains	16
itr_{max}	Maximum number of iterations in WLS	40

size of the salient domains grows larger. Not surprisingly, this phenomenon can be attributed to the declining accuracy of the significant domain clustering as the size of the significant domain increases. Furthermore, when the value of S is less than 16, the completeness of the fused DSM no longer exhibits significant improvement, and the RMSE no longer decreases drastically. Consequently, the inflection point $S = 16$ for salient domain clustering also serves as the inflection point for the change in the fused DSM. Therefore, for a balance between computational complexity and accuracy of the fused results, the value of S_{opt} selected for salient domain clustering is determined to be 16.

3) Sensitivity Experiments on the Number of Iterations: Based on the results of previous sensitivity experiments, the remaining parameters utilized are delineated in Table II. Compl. and RMSE of DSMs from different four test areas are compared by varying the number of iterations. The curve diagrams of deviation value, Compl., and RMSE corresponding to different numbers of iterations are shown in Fig. 12.

Based on Fig. 12, it can be observed that, overall, the deviation values in elevation of the fused DSMs for the four areas exhibit a decreasing trend as the number of iterations increases, while the completeness demonstrates an increasing trend, and the RMSE shows a decreasing trend. Furthermore, when the number of iterations surpasses 40, there is no substantial alteration in the aforementioned three indicators, signifying a diminishing effect of iteration on the optimization of DSM fusion. Taking into account the computational complexity associated with the number of iterations, the optimal number of iterations is, therefore, determined to be 40.

In conclusion, the sensitivity analysis of significant parameters indicates that for subsequent qualitative and quantitative comparative experiments, the values of the parameters of interest should follow those delineated in Table III.

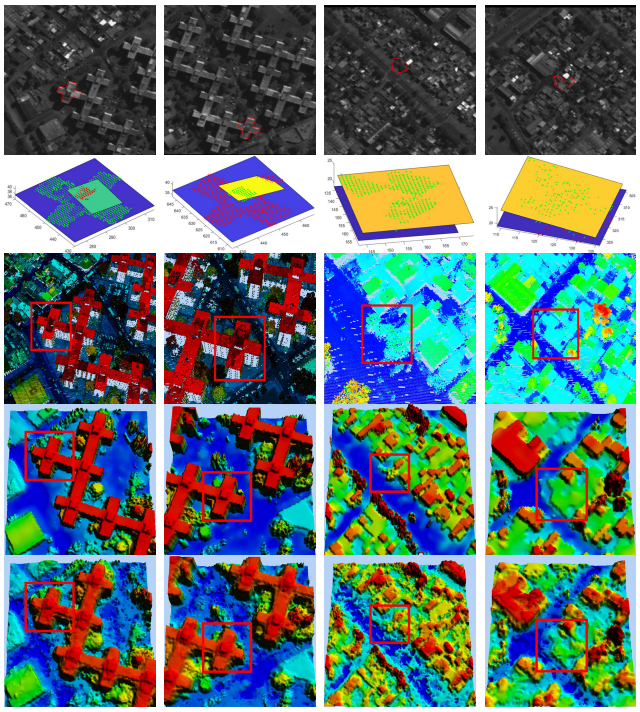
B. Evaluation About the Effectiveness of CDC for DSM Fusion

1) Effects of Salient Domain Clustering on DSM Fusion: In the experiments investigating the efficacy of DSM fusion, we tracked the results of multiview DSM fusion in four distinct areas, namely, complex buildings, trees, and low buildings from Fig. 10, as illustrated in Fig. 13.

From Fig. 13(a) and (b), it is evident that salient domain clustering can only effectively differentiate walls with clear edges, but struggles to identify raised structures on the roof with weak textures. After that, the classes of detailed structures are clustered by the model domain clustering, and then, the purpose of fine structure segmentation is achieved [see row 2 of Fig. 13(a) and (b)]. As such, the multiview DSM fusion is capable of being constrained by the building edges and the geometric models [see rows 3–5 of Fig. 13(a) and (b)]. Although fine surface structures are uncovered by model domain clustering, the role of salient domain clustering cannot be ignored. The reason is that, salient domain clustering is bale to delineate clear target edges for model domain clustering [see row 2 of Fig. 13(a) and (b)]. More importantly, salient regions allow the number of potential categories to be reduced through the aggregation effect of similar pixels, enhancing the accuracy of clustering and reducing the time spent on clustering.

In regard to tree clustering, as depicted in Fig. 13(c), it can be observed that salient domain clustering only manages to capture a portion of the tree canopy, which also includes a fraction of the ground. Consequently, this leads to errors in the fused DSM pertaining to the canopy. On the other hand,

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(a) (b) (c) (d)

Fig. 13. Outcomes of salient domain clustering and the corresponding results of multiview DSM fusion. (a)–(d) Complex buildings, complex buildings, trees, and low buildings, respectively. Rows 1–5 display the following: salient regions, geometric modes, LiDAR point clouds, outcomes of the proposed algorithm, and outcomes of direct median filtering.

Fig. 13(d) showcases the outcomes for low buildings, where the close proximity of neighboring buildings hampers the ability of direct median filtering to effectively discern between different structures. In contrast, the DSM fused with salient domain clustering exhibits clearer, more precise building edges that are in greater alignment with the actual LiDAR point cloud, compared to the DSM fused without the incorporation of supplementary constraints.

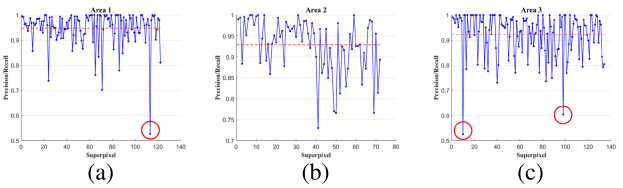
2) Evaluation of Model Domain Clustering: As the surface categories within each salient region are different and the number of categories is also different, it is not significant to calculate *Recall* or *Precision* for only a single category. This section evaluates the multicategory accuracy by calculating multicategory Precision, Recall, and micro-F1 with the following equations:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{36}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{37}$$

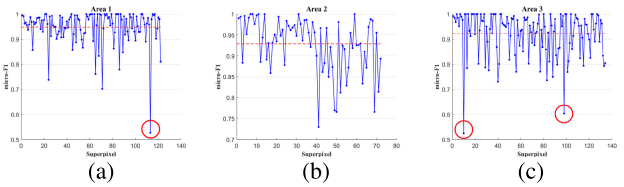
$$\text{micro-F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Preciosion} + \text{Recall}} \tag{38}$$

where the current target category is a *Positive* sample and the other categories are *Negative* samples in this experiment. TP denotes the sum of *True Positive* samples for all categories, FP denotes the sum of *False Positive* samples for all categories, and FN denotes the sum of *False Negative* samples for all categories. In the multiclassification task, FP = FN and Precision = Recall; thus, Precision = Recall = micro-F1.



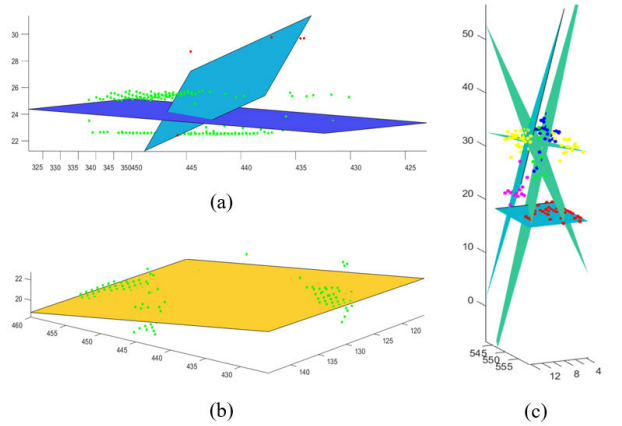
(a) (b) (c)

Fig. 14. Precision/recall evaluation results. (a)–(c) Evaluation results of three test areas, and the red dotted lines indicate the average values.



(a) (b) (c)

Fig. 15. Micro-F1 evaluation results. (a)–(c) Evaluation results of three test areas, and the red dotted lines indicate the average values.



(a) (b) (c)

Fig. 16. Model domain clustering results. The 3-D points colored differently indicate different categories. (a)–(c) Results of model domain clustering corresponding to the three points marked by the red circles in Fig. 15.

To sufficiently evaluate the accuracy of model domain clustering and to verify its applicability to stereo DSM data, we evaluate the clustering algorithm in three different areas, which have 1219, 718, and 1340 salient regions, respectively. The clustering accuracy of some samples is then obtained by uniform sampling with the interval of 10, and plotted as the line plots. The statistical results are shown in Figs. 14 and 15.

Based on the analysis presented in Figs. 14 and 15, the average Precision, Recall, and micro-F1 values exceed 90%. Nevertheless, it is worth noting that there exist three salient regions within the test areas where the clustering results exhibit accuracies below 70%, as indicated by the red circles in Figs. 14 and 15. Consequently, we conducted a comprehensive examination of the clustering outcomes characterized by low accuracies. Subsequently, the clustering results corresponding to the three points marked by the red circles in Fig. 14 are illustrated in Fig. 16.

Based on Fig. 16, the errors existing in the model domain clustering can be divided into two types. The first type is that 3-D points located in different regions but with the same geometric structure are clustered into one class, such as the green points in Fig. 16(a) and (b). The second type

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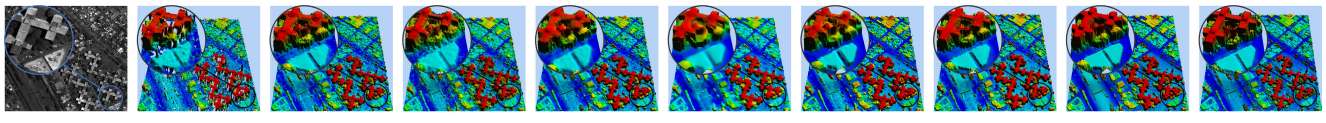


Fig. 17. Qualitative evaluation results in Area 1. (a) Panchromatic image. (b) Best selected input DSM. (c) DSM of PVR15. (d) DSM of MFM15. (e) DSM of AKC17. (f) DSM of AMF19. (g) DSM of UGD22. (h) DSM of AIG22. (i) DSM of IBI23. (j) DSM of the proposed algorithm.

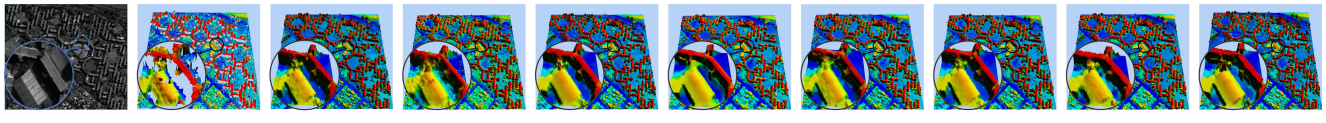


Fig. 18. Qualitative evaluation results in Area 2. (a) Panchromatic image. (b) Best selected input DSM. (c) DSM of PVR15. (d) DSM of MFM15. (e) DSM of AKC17. (f) DSM of AMF19. (g) DSM of UGD22. (h) DSM of AIG22. (i) DSM of IBI23. (j) DSM of the proposed algorithm.

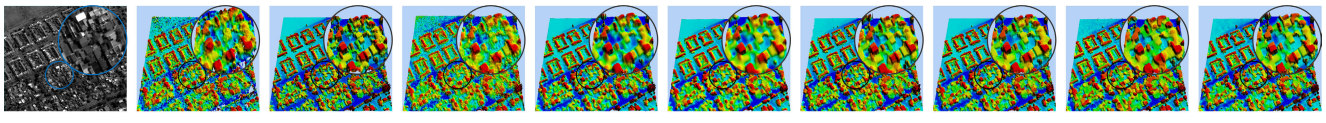


Fig. 19. Qualitative evaluation results in Area 3. (a) Panchromatic image. (b) Best selected input DSM. (c) DSM of PVR15. (d) DSM of MFM15. (e) DSM of AKC17. (f) DSM of AMF19. (g) DSM of UGD22. (h) DSM of AIG22. (i) DSM of IBI23. (j) DSM of the proposed algorithm.

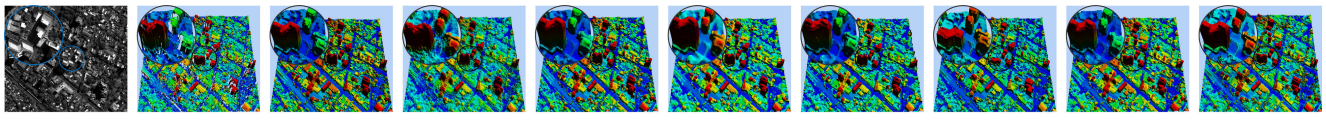


Fig. 20. Qualitative evaluation results in Area 4. (a) Panchromatic image. (b) Best selected input DSM. (c) DSM of PVR15. (d) DSM of MFM15. (e) DSM of AKC17. (f) DSM of AMF19. (g) DSM of UGD22. (h) DSM of AIG22. (i) DSM of IBI23. (j) DSM of the proposed algorithm.

is that the 3-D points with identical geometric structures are excessively fragmented and assigned to disparate categories due to over-clustering, such as the yellow, blue, and green points in Fig. 16(c). It is not likely that the first type of error can influence the effectiveness of the filtering from geometric model since the geometric model with the same surface structure within a single salient region has a consistent filtering effect on outliers. In addition, the purpose of model domain clustering is to divide different topographic areas within a single salient region, and the finer the geometric model is, the more it can play the role of filtering error points, so the second type will not bring impact on the fusion of multiview DSM instead.

Finally, we also find that a few outlier 3-D points originating from the stereo DSM are clustered into one class, such as the red 3-D points in Fig. 16(b). While this category cannot be classified as a clustering error in terms of evaluating clustering accuracy, as the class to which the outlier points belong is indeed distinguished, the outlier class tends to misguide the fusion of multiview DSMs. Nevertheless, this error is able to be iteratively optimized by WLS, as detailed in Section III-C.

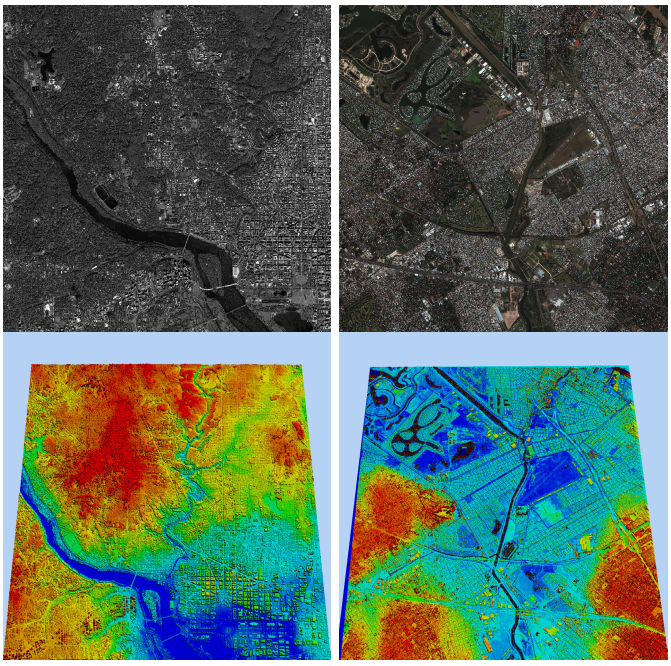
C. Evaluation of Multiview DSM Fusion

1) Qualitative Comparison Experiments: The popular multiview DSM fusion methods used for qualitative comparison include the PVR15 in [13], the mean filtering method (denoted as MFM15, which is an improved method based on PVR15 by replacing pointwise median calculation with the mean calculation) [13], the AKC17 in [2], the AMF19 in [14], the UGD22 in [15], the AIG22 in [16], and the IBI23 in [12], and these DSMs are available.[2] The results of Areas 1–4 in Fig. 10 are shown in Figs. 17–20, respectively.

In general, apart from the MFM15 algorithm, the other algorithms are able to optimize the stereo DSM and achieve visually satisfactory results according to Figs. 17–20. For PVR15, large buildings can be recovered [see Fig. 18(c)], but the protection of details is not sufficient such as raised patches on building roofs. And there are small errors in areas with smooth surface and elevation drop-offs [see Figs. 17(c) and 20(c)]. In addition, the shape of small buildings is deformed and distorted, and errors exist at the edges [see Fig. 19(c)]. Compared with PVR15, AKC17 can recover the shapes of small buildings [see Fig. 19(e)] and obtain accurate building edges [see Figs. 18(e) and 20(e)]. However, there are lots of salt-and-pepper noises in the smooth regions [see Fig. 18(e)] and the regions with severe elevation drop [see Fig. 17(e)]. AMF19 tackles the problem of smooth surface noise [see Fig. 18(f)], but the DSM obtained by AMF19 is disturbed by noises at the edges of buildings, has blurred building edges, and suffers from severe loss of

[2]https://github.com/zhao1i/CDC-qualitative-comparison-experiments-DSM.git

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(a) (b)

Fig. 21. DSM fusion results for WV-2 and WV-3 images. (a) DSM of WV-2. (b) DSM of WV-3.

detailed surface structures [see Figs. 17(f) and 20(f)]. The UGD22 algorithm introduces a novel biclassification approach, utilizing matching confidence in conjunction with median filtering. Consequently, the overall performance of UGD22 surpasses that of PVR15. Nevertheless, it is noteworthy that the confidence biclassification does not fully address the issue of deformation in fine building structures [see Fig. 17(g)]. The AIG22 algorithm employs in-window elevation bisection to derive the median elevation within a specific range, thereby safeguarding the boundaries of clustered buildings based on elevation disparities. However, this approach leads to excessive smoothing of intricate building structures [see Figs. 17(h) and 20(h)]. The essence of the IBI23 algorithm lies in optimizing the DSM through iterative bilateral integration and difference shifting. However, a challenge arises as the correct elevations may be misaligned during the process of shifting the complete stereo DSM based on the difference, resulting in blind DSM migration and a lack of orientation [see Figs. 17(i) and 20(i)]. Conversely, it is found that the proposed algorithm not only restores the surface details of low-rise buildings [see Fig. 17(j)], but it also sharpens the edges of high-rise buildings [see Fig. 20(j)]. Furthermore, the proposed algorithm also reconstructs the curved roof structure of the domed building [see Fig. 18(j)].

In order to validate the broad applicability of the proposed algorithm, it was implemented on entire WV-2 and WV-3 imageries, with dimensions of 14 780 × 14 524 and 27 422 × 26 701, respectively, as illustrated in Fig. 21.

2) Quantitative Comparison Experiments: In this section, the DSMs obtained by the proposed algorithm are quantitatively assessed based on the WV-3 images with ground-truth

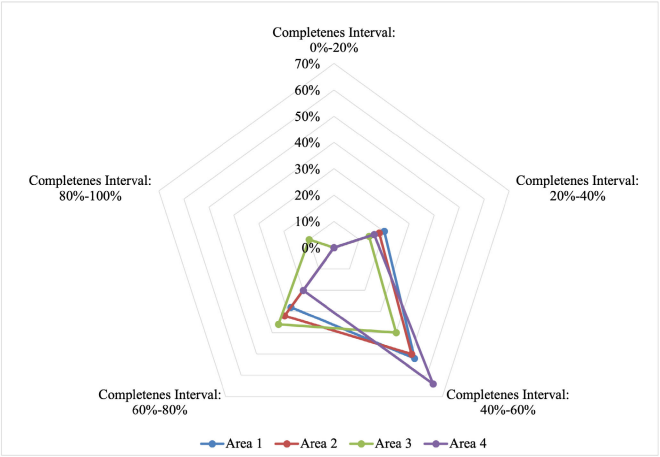


Fig. 22. Distribution of stereo DSMs completeness for four test areas.

LiDAR point clouds. However, the robustness of the multiview fusion algorithm cannot be adequately assessed if the stereo DSMs are not considered. It is clear that the completeness and accuracy of the stereo DSMs utilized for fusion significantly impact the fusion algorithm. For this reason, we assessed the completeness of all the stereo DSMs in the four test areas, specifically 50 stereo DSMs per test area, amounting to a total of 200 DSMs. To gain a more lucid perception of the effectiveness of the stereo DSMs utilized for fusion, the completeness was initially divided into five intervals: 0%–20%, 20%–40%, 40%–60%, 60%–80%, and 80%–100%. Then, the percentage of stereo DSMs corresponding to the aforementioned five intervals is presented in Fig. 22.

According to Fig. 22, it can be found that the completeness of the stereo DSM varies across different regions. In Area 1, Area 2, and Area 4, the completeness is predominantly within the interval of 40%–60%. In Area 3, however, the completeness is distributed within the intervals of 40%–60% as well as 60%–80%. It is worth noting that only a small proportion of stereo DSMs in Area 3 exhibit completeness exceeding 80%. Furthermore, the completeness of the stereo DSM is most optimal in Area 3, characterized by a dense distribution of low-rise buildings, and least optimal in Area 4, characterized by a dense distribution of high-rise buildings.

Moreover, for a more comprehensive evaluation of the fused DSM quality, in addition to RMSE and Compl., the mean absolute percentage error (MAPE) and the normalized median absolute residuals (NMARs) as proposed by Rothermel et al. [44] should be considered. These evaluation metrics are calculated as follows.

1) MAPE is the ratio of the absolute value of elevation error to the true elevation value; thus, MAPE can quantify the degree of deviation from the true elevation

$$\text{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{E_i - \text{Tru}_i}{\text{Tru}_i}\right| \times 100\%. \tag{39}$$

2) NMAR is essentially the median of the absolute values of the elevation errors, which is able to be resistant to

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TABLE IV
QUANTITATIVE EVALUATION RESULTS IN AREA 1

	INPUT	PVR15	MFM15	AKC17	AMF19	UGD22	AIG22	IBI23	Prop
RMSE(m)	2.69	1.88	2.03	1.87	2.09	2.12	1.59	1.86	**1.00**
MAPE(%)	8.34	8.81	9.14	6.32	7.08	6.26	5.37	7.39	**5.25**
Compl.(%)	66.74	77.27	64.59	84.22	76.95	79.43	85.32	84.78	**92.74**
NMAR(m)	2.51	1.39	1.81	1.04	1.24	**0.89**	0.92	1.09	1.07
Time(min)	/	**4.32**	5.57	99.23	15.38	21.12	185.33	12.95	19.08

TABLE V
QUANTITATIVE EVALUATION RESULTS IN AREA 2

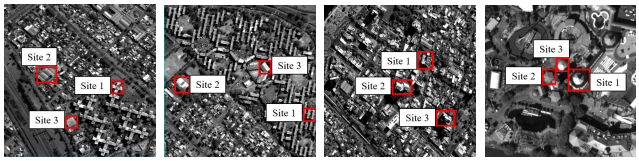
	INPUT	PVR15	MFM15	AKC17	AMF19	UGD22	AIG22	IBI23	Prop
RMSE(m)	2.43	0.74	1.31	0.75	1.48	1.55	1.12	1.49	**0.65**
MAPE(%)	8.88	4.54	6.70	4.58	5.71	5.74	4.57	6.84	**4.35**
Compl.(%)	69.53	89.70	75.31	91.78	78.48	77.48	83.96	81.56	**92.12**
NMAR(m)	4.70	1.47	1.66	1.09	1.41	1.42	0.87	1.32	**0.80**
Time(min)	/	**4.07**	9.40	91.6	5.07	23.36	176.87	8.07	20.17

TABLE VI
QUANTITATIVE EVALUATION RESULTS IN AREA 3

	INPUT	PVR15	MFM15	AKC17	AMF19	UGD22	AIG22	IBI23	Prop
RMSE(m)	1.39	0.59	0.87	**0.50**	0.84	0.97	0.72	0.94	**0.50**
MAPE(%)	4.93	4.11	5.65	3.45	4.22	4.48	3.51	5.32	**3.39**
Compl.(%)	80.17	91.22	81.71	90.49	86.12	83.70	88.19	86.89	**91.54**
NMAR(m)	1.94	0.67	1.14	0.57	0.69	0.70	0.64	0.90	**0.56**
Time(min)	/	2.17	2.67	47.08	4.02	14.89	171.33	5.28	11.81

TABLE VII
QUANTITATIVE EVALUATION RESULTS IN AREA 4

	INPUT	PVR15	MFM15	AKC17	AMF19	UGD22	AIG22	IBI23	Prop
RMSE(m)	4.30	2.56	2.87	2.64	2.81	3.24	3.30	2.89	**1.55**
MAPE(%)	10.11	7.80	10.21	6.27	7.79	8.55	6.42	8.81	**5.20**
Compl.(%)	63.51	82.89	64.55	84.58	74.96	73.05	81.33	78.21	**89.65**
NMAR(m)	3.64	1.60	1.83	**0.77**	1.16	1.21	1.80	1.14	0.92
Time(min)	/	**5.25**	6.97	92.4	12.22	22.37	191.29	16.65	20.08



(a) (b) (c) (d)

Fig. 23. Distribution of test sites. (a)–(d) Test sites are selected from 4 different test areas.

extreme errors and reflect the general magnitude of the error

$$\mathrm{NMAR} = \mathrm{Med}\left(\left\{\mathrm{err}_i \,\middle|\, \begin{matrix}\mathrm{err}_i = |E_i - \mathrm{Tru}_i|, \\ i = 1, \ldots, N\end{matrix}\right\}\right) \quad (40)$$

where E_i and Tru_i denote the elevation value and the ground-truth value at the ith nonempty cell, respectively. N is the number of nonempty cells. $\mathrm{Med}(\cdot)$ is the median function.

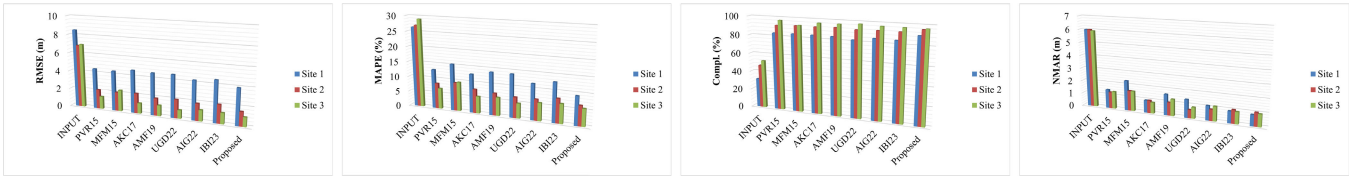
a) Comparison of DSM in mixed areas: The evaluation results about these fusion algorithms in Areas 1–4 are presented in Tables IV–VII, respectively. The four areas contain complex scene information, including buildings, vegetations, roads, and vehicles, which puts higher requirements on the robustness of the fusion algorithms.

TABLE VIII
COMPARATIVE ASSESSMENT BETWEEN MIXED ZONES AND BUILT-UP SITES (THE OPTIMAL VALUES ARE BOLDED)

	Ave. RMSE (m)	Ave. MAPE (%)	Ave. Compl. (%)	Ave. NMAR (m)
Mixed areas	**0.93**	**4.55**	91.51	**0.84**
Built-up areas	3.00	7.24	**93.10**	0.87

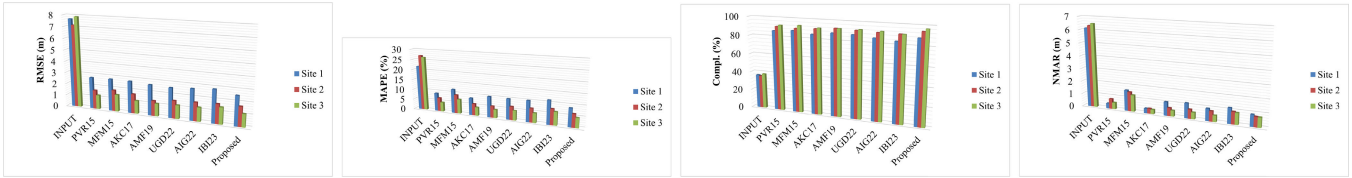
As shown in Tables IV–VII, the optimal values are bolded, and the DSMs obtained by all the fusion algorithms outperform the original input DSM in terms of completeness and accuracy. Obviously, the proposed algorithm outperforms the currently popular multiview DSM fusion algorithms in terms of accuracy and completeness. Moreover, in terms of the evaluation metrics RMSE, MAPE, Compl., and NMAR, the proposed algorithm achieved average values of 0.93 m, 4.55%, 91.51%, and 0.84 m, respectively, across the four distinct regions. However, due to the fact that the proposed method requires plenty of time for model domain clustering, and the time spent for model domain clustering is about 15 min.

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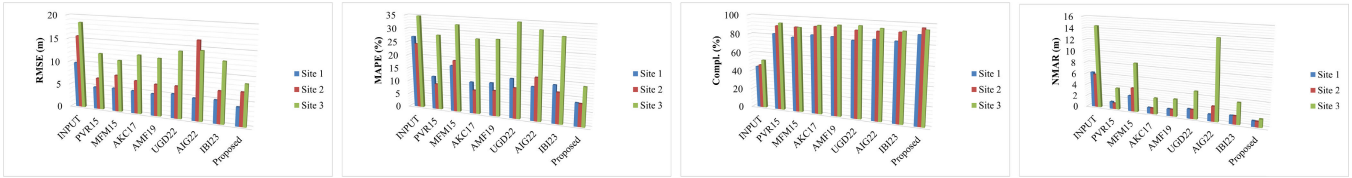
(a) (b) (c) (d)

Fig. 24. Quantitative evaluation results about the three test sites in Fig. 23(a). (a)–(d) Values of the metrics RMSE(m), MAPE(%), Compl.(%), and NMAR(m), respectively.



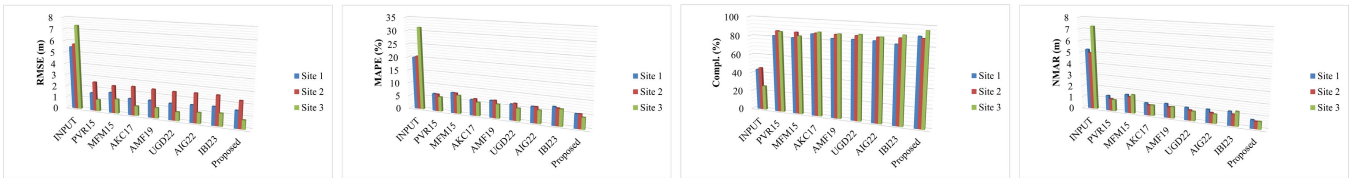
(a) (b) (c) (d)

Fig. 25. Quantitative evaluation results about the three test sites in Fig. 23(b). (a)–(d) Values of the metrics RMSE(m), MAPE(%), Compl.(%), and NMAR(m), respectively.



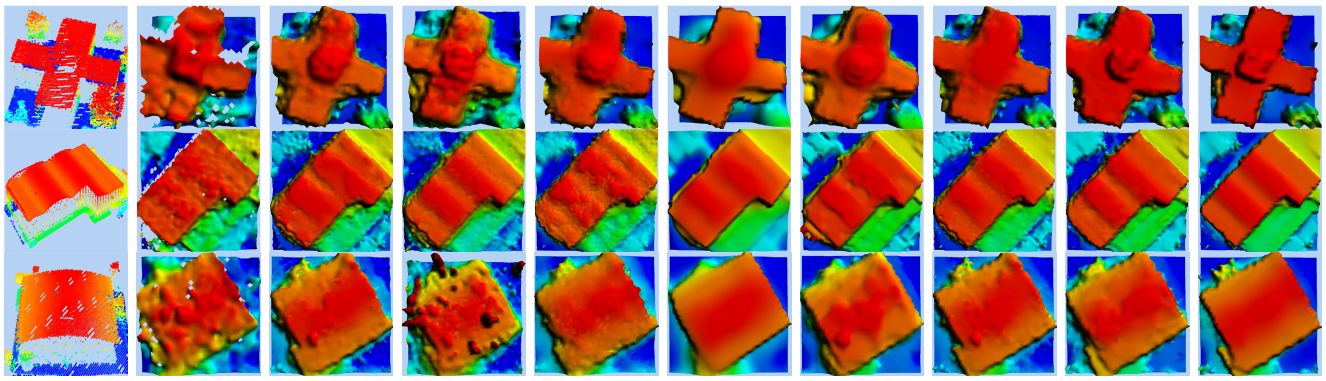
(a) (b) (c) (d)

Fig. 26. Quantitative evaluation results about the three test sites in Fig. 23(c). (a)–(d) Values of the metrics RMSE(m), MAPE(%), Compl.(%), and NMAR(m), respectively.



(a) (b) (c) (d)

Fig. 27. Quantitative evaluation results about the three test sites in Fig. 23(d). (a)–(d) Values of the metrics RMSE(m), MAPE(%), Compl.(%), and NMAR(m), respectively.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 28. DSMs of the three test sites in Fig. 23(a). (a) Ground-truth LiDAR point cloud. (b) INPUT. (c) PVR15. (d) MFM15. (e) AKC17. (f) AMF19. (g) UGD22. (h) AIG22. (i) IBI23. (j) Proposed. Rows 1–3 show the results of the three test sites.

b) Comparison of DSM in built-up areas: It is hypothesized that vegetated areas and physical changes in the ground interfere with the completeness of the DSM. For that, three test sites that merely contain the buildings, roads, and ground are selected from four areas for quantitative evaluation, as shown in Fig. 23. These test sites include stand-alone mid-rise buildings [see the sites in Fig. 23(a)], dense mid-rise buildings [see the sites in Fig. 23(b)], stand-alone high-rise buildings [see the sites in Fig. 22(c)], and cone-roofed buildings [see the sites in Fig. 22(d)]. The quantitative assessment outcomes in terms of

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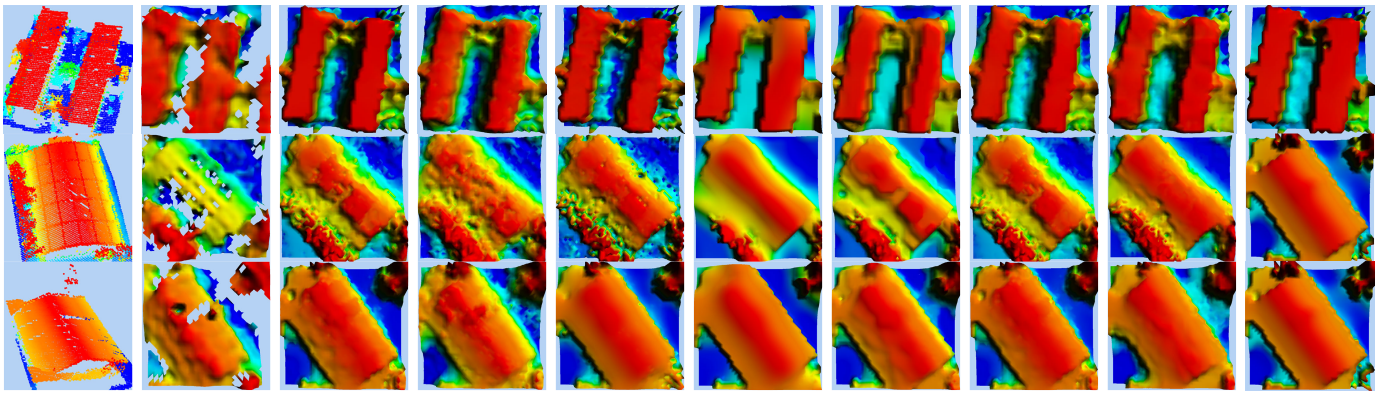


Fig. 29. DSMs of the three test sites in Fig. 23(b). (a) Ground-truth LiDAR point cloud. (b) INPUT. (c) PVR15. (d) MFM15. (e) AKC17. (f) AMF19. (g) UGD22. (h) AIG22. (i) IBI23. (j) Proposed. Rows 1–3 show the results of the three test sites.

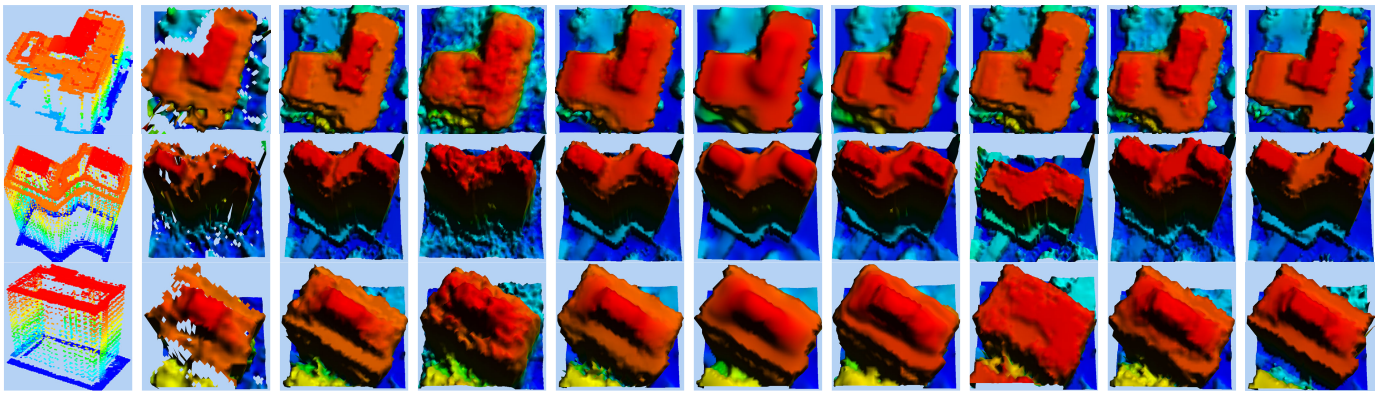


Fig. 30. DSMs of the three test sites in Fig. 23(c). (a) Ground-truth LiDAR point cloud. (b) INPUT. (c) PVR15. (d) MFM15. (e) AKC17. (f) AMF19. (g) UGD22. (h) AIG22. (i) IBI23. (j) Proposed. Rows 1–3 show the results of the three test sites.

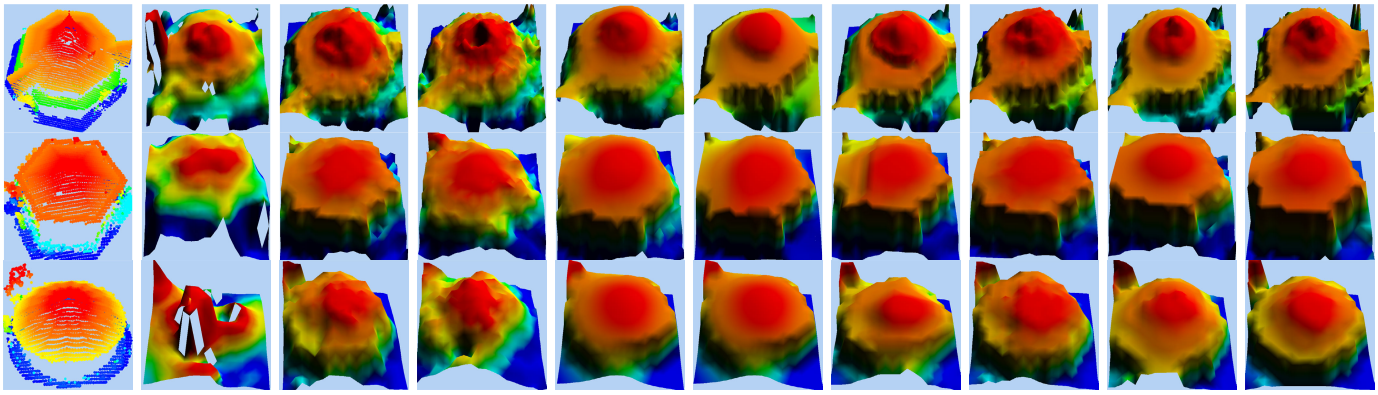


Fig. 31. DSMs of the three test sites in Fig. 23(d). (a) Ground-truth LiDAR point cloud. (b) INPUT. (c) PVR15. (d) MFM15. (e) AKC17. (f) AMF19. (g) UGD22. (h) AIG22. (i) IBI23. (j) Proposed. Rows 1–3 show the results of the three test sites.

RMSE, MAPE, Compl. and NMAR across the 3 $\times$ 4 test sites are delineated in Figs. 24–27, with the corresponding DSMs illustrated in Figs. 28–31.

According to Figs. 24–27, in general, the DSM in the building area possesses better completeness compared to the mixed area. Moreover, the proposed algorithm has the best results and MFM15 has worst results among these fusion algorithms in terms of accuracy, completeness, and error deviation in most of the test sites. For specificity, among all the test sites in the four areas, site 3 of Area 3 (see Fig. 26), where the high-rise buildings are situated, is where our algorithm demonstrates the highest superiority compared to the other fusion algorithms. In particular, the proposed algorithm exceeds other fusion algorithms by about one-third for the metrics RMSE and

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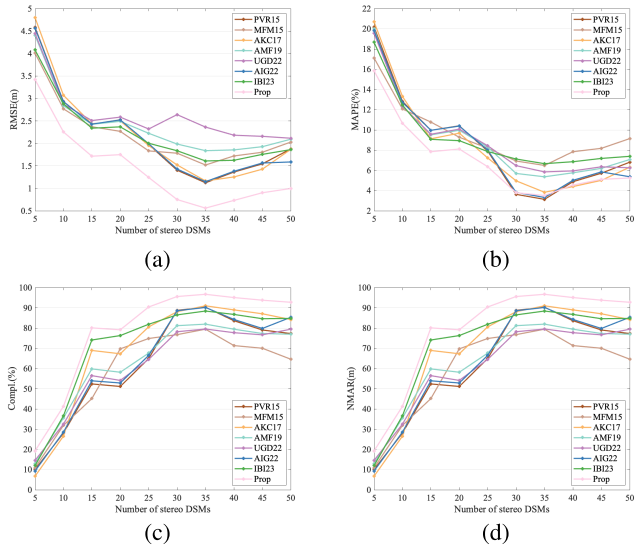


Fig. 32. Robustness assessment outcomes utilizing variable quantities of stereo DSMs. (a)–(d) Evaluation results for the metrics of RMSE, MAPE, Compl., and NMAR, respectively.

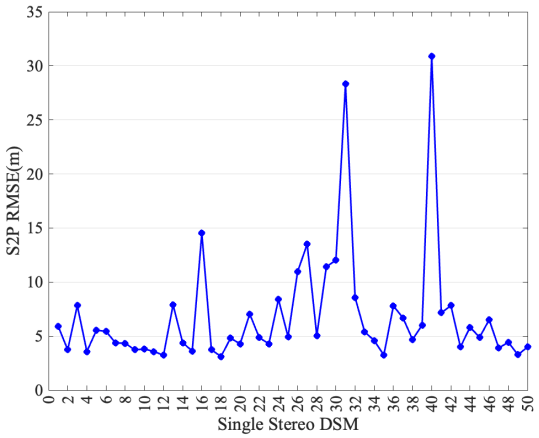


Fig. 33. Evaluation results of stereo DSMs from S2P.

MAPE, which reflect the degree of elevation differences. It is thanks to the interaction of salient domain clustering and model domain clustering in the proposed algorithm that the final DSM enables to obtain accurate building edges with severe elevation drop-offs.

As seen from Figs. 28–31, PVR15, AKC17, AMF19, UGD22, AIG22, IBI23, and the proposed algorithm are generally able to recover the surface structures of the buildings. To be specific, in terms of the fineness of the building edges, the proposed algorithm outperforms the other algorithms, which is attributed to the constraint of class regions from model domain clustering. Moreover, compared with the ground-truth LiDAR point cloud, the proposed algorithm reconstructs the dome-shaped and cone-shaped structures well through WLS [see Figs. 29(j) and 31(j)].

In addition, to compare mixed areas and built-up areas, Table VIII counts the mean values of RMSE, Compl., and NMAR about the proposed algorithm between the four mixed areas and the 12 built-up sites.

According to Table VIII, the built-up areas outperforms the mixed areas in terms of indicator Compl. It stands to reason that error points situated along building boundaries are inclined to exhibit greater deviations relative to other regions, thereby resulting in higher RMSE, MAPE, and NMAR.

c) Robustness comparison with popular algorithms: In order to ascertain the robustness of the various algorithms with respect to the number of stereo DSMs, assessments were conducted using ten different DSM counts, specifically 5, 10, 15, 20, 25, 30, 35, 40, 45, and 50. Moreover, the complex geomorphological landscape of Area 1, as delineated in Fig. 10, was selected to serve as the test area for analysis, as shown in Fig. 32.

Based on the findings depicted in Fig. 32, the proposed algorithm demonstrates superiority over other widely used fusion algorithms in terms of DSM completeness and accuracy. Furthermore, the advantage of the proposed algorithm remains consistent even as the number of stereo DSMs used for fusion varies. Nevertheless, with an increase in the number of stereo DSMs, the Compl. of fused DSMs does not consistently improve, and the RMSE does not consistently decrease; rather, it fluctuates [see Fig. 32(a) and (c)]. Specifically, there are inverse fluctuations in the Compl. and RMSE of the fused DSMs within the intervals of 15–20 and 35–45. To analyze this phenomenon, we assessed the RMSE of all the DSMs generated by the S2P pipeline within Area 1, ensuring that the order of the stereo DSMs remained unchanged. The results of this assessment are presented in Fig. 33.

Fig. 33 highlights the presence of stereo DSMs with significant RMSE values within the 15–20 and 35–50 intervals, respectively. Specifically, the 16th, 31st, and 40th stereo DSMs exhibit high RMSE values. Clearly, errors in the stereo DSMs have a direct impact on multiview DSM fusion. Nevertheless, compared to other widely-used method, the proposed algorithm exhibits superior effectiveness in mitigating the influence of inaccurate 3-D points. Specifically, considering the average completeness and RMSE of the stereo DSM across the four mixed areas, the proposed algorithm enhances the completeness of the stereo DSM from 50.87% to 91.51%, representing a 79.89% improvement, and decreases the RMSE from 6.84 m to 0.93 m, indicating a 86.40% reduction.

V. Conclusion

To optimize the completeness and accuracy of DSMs from multiview urban satellite images, we have proposed a novel multiview DSM fusion method based on CDC. Specifically, a coarse-to-fine strategy is adopted, where the elevation divisions with sharp elevation drop are first enhanced under the constraint of salient region edges, and then, the fine surface structures hidden in weakly textured regions are rebuilt guided by geometric models. Eventually, the accurate DSMs are achieved by iterative optimization of WLS with the class regions as the units.

The evaluation experiments on the multiple stereo DSMs generated from WV-2 and WV-3 urban satellite images show that not only the elevation divisions of objects with sharp elevation drop can be enhanced (e.g., building edges), but also the fine surface structures are able to be restored (e.g.,

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raised water towers on the building roofs). Compared with the ground-truth LiDAR point cloud, the average RMSE of the DSM obtained by the proposed algorithm is 0.93 m, with 91.51% completeness. In comparison to the stereo DSMs, the fused DSMs generated by the proposed algorithm exhibit, on average, an 86.40% reduction in RMSE and a 79.89% increase in completeness. Additionally, the proposed method surpasses other popular fusion methods in preserving detailed surface structures, reconstructing cone-roofed buildings, and maintaining robustness when using imperfect stereo DSMs.

Unfortunately, fusing DSMs of vegetation areas with significant seasonal differences and suppressing elevation errors at the edges of high-rise buildings remain challenging problems to solve. As the height of vegetation varies seasonally, it is difficult to distinguish the accuracy of changing heights over time using fusion algorithms alone, without considering the time factor. It is probably a new research hotspot in the field of 3-D reconstruction based on satellite images combining satellite imagery-based change detection. Such that, changes in elevation are first identified through analysis of multitemporal satellite images. DSM fusion is then performed within areas of no change, and the elevations of the changed areas are replaced according to time. Furthermore, detecting individual buildings enables reconstructing 3-D models of high-rise structures, offering additional promise for representing urban morphology.

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